Biochemical and Biophysical Research Communications 762 (2025) 151749

Contents lists available at ScienceDirect

Biochemical and Biophysical Research Communications

journal homepage: www.elsevier.com/locate/ybbrc

# *MGAT3* and *MGAT5* overexpression alters the protein cargo of extracellular vesicles released by metastatic melanoma cells

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ARTICLE INFO

*Keywords:*
Biomarkers
Extracellular vesicles
Glycosylation
Melanoma
Proteomics

ABSTRACT

Extracellular vesicles (EVs) are potential non-invasive diagnostic, prognostic and therapeutic tools. Additionally, they are important contributors to tumorigenesis. Glycosylation has been found to modulate the composition of the EV proteome. Increased amounts of β1,6-branched N-glycans, synthesized by N-acetylglucosaminyltransferase V (GnT-V), are most commonly observed in melanoma and are associated with decreased cell adhesion and increased metastasis. The opposite effect is caused by the addition of bisecting GlcNAc by N-acetylglucosaminyltransferase III (GnT-III). To date, the impact of these enzymes on EV cargo in melanoma remains unexplored. Flow cytometry was used to study the surface glycosylation of genetic variants of WM266-4 melanoma cells with induced overexpression of GnT-III or GnT-V encoding genes (*MGAT3* or *MGAT5*) and EVs released by these cells.

LC-MS/MS proteomics was applied to analyze the effect of altered glycosylation on the proteome of released EVs, followed by detailed bioinformatic analysis. Flow cytometry analysis revealed dynamic changes in the surface glycosylation of EVs derived from melanoma cells overexpressing *MGAT3* or *MGAT5*. Induced overexpression of *MGAT3* or *MGAT5* also caused significant changes in the proteome of EVs. The proteomic analysis identified a total of 1770 microvesicular and 704 exosomal proteins that play different roles in melanoma progression, including those with established diagnostic/prognostic potential and those closely associated with melanoma onset. Proteomic profiling of EVs derived from cells overexpressing *MGAT3* and *MGAT5* revealed functional changes in EV protein content driven by glycosylation modifications. The study presented a potential multifaced application of melanoma-derived EVs for diagnostic and prognostic purposes.

## 1. Introduction

Malignant melanoma, although less common than non-melanoma skin cancers, is the leading cause of skin cancer deaths. Moreover, the incidence and prevalence rates of cutaneous melanoma continue to rise [1]. Therefore, there is an urgent need for early detection of melanoma, which is essential for effective treatment. However, achieving this through histopathological diagnosis remains a major challenge due to the lack of reliable markers.

Current melanoma treatment strategies include a combination of surgery, medical therapies and radiation therapy. The first step in treatment is usually surgical removal of the melanoma with a margin of healthy tissue. For advanced-stage melanomas, immunotherapy with checkpoint inhibitors (such as pembrolizumab or nivolumab, which block PD-1, PD-L1, or CTLA-4) [2–4], cytokine therapy (e.g., IL-2) [5], or FDA-approved vaccines such as Imlygic [6] are commonly used. Since *BRAF* and *MEK* are frequently mutated in melanoma, BRAF inhibitors (e.g., vemurafenib, dabrafenib) [7–9] and MEK inhibitors (e.g., trametinib) [10] are often used, especially in patients with the $BRAF^{V600}$ mutation. In cases where other treatments are unsuccessful, chemotherapy may be considered. Radiation therapy is also used, especially when melanoma has spread to the brain or other areas where surgery is not possible. It is often employed in palliative care to relieve symptoms and control tumor growth.

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https://doi.org/10.1016/j.bbrc.2025.151749
Received 12 December 2024; Received in revised form 25 March 2025; Accepted 1 April 2025
Available online 2 April 2025
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Nevertheless, studies over the past decades have shown that tumor-derived extracellular vesicles (EVs) can be successfully used for non-invasive diagnosing, monitoring and treatment of cancer patients [11–16], including patients with melanoma [17–19]. EVs are nanoscale, membrane-bound vesicles released into the extracellular space and can be detected in all body fluids that are key mediators in cell-to-cell communication. Based on their biogenesis mechanism, size, and the presence of specific markers, EVs are divided into exosomes (EXOs), microvesicles (MVs, also known as ectosomes) and apoptotic bodies. The involvement of EVs in supporting tumor cell growth and survival, promoting tumor metastasis and angiogenesis, reprogramming the tumor environment, development of immunosuppression, and drug resistance is well documented [20–24], also in the case of melanoma [17,25,26]. This is achieved by horizontal transfer of a number of bioactive molecules, including proteins, lipids and nucleic acids, etc., through EVs from parental into recipient cells.

For a very long time, most studies on the use of components of EV cargo for diagnostic or therapeutic purposes focused on RNAs. However, tumor-derived EV proteins also show this potential and are an important contributor to tumorigenesis [27]. Therefore, studying the protein cargo in melanoma-derived EVs may provide deeper insights into the mechanisms driving melanoma progression. Proteins present in EVs can be as well used to identify potential diagnostic and/or predictive biomarkers and treatment targets. Nevertheless, the amount of data on this topic is significantly limited [28]. Analysis of the melanoma-derived EV proteome based on results obtained from *in vitro* studies and melanoma patient samples is promising, but their respective clinical value requires validation in multicenter trials.

Melanoma cells have been shown to release both EXOs and MVs [29]. In our previous *in vitro* studies on EVs derived from human melanoma cells, we have demonstrated the presence of proteins that are routinely used as melanoma markers in clinical practice, as well as other proteins whose feasibility as potential biomarkers has not yet been confirmed in melanoma patients [30,31]. In addition, we showed that the percentage of proteins assigned to major cancer-related categories corresponded to the more aggressive phenotype of melanoma cells releasing them. In other studies, we have shown that melanoma progression is accompanied by changes in the molecular composition of the EVs they release, including in particular the secondary structure of proteins, that can be used as markers of metastasis development [32]. Additionally, since changes in glycosylation, which is one of the most common post-translational modifications (PTMs) of proteins, are hallmarks of cancer, we also analyzed glycosylation patterns on the surface of melanoma-derived EVs by flow cytometry [33]. We found that the amount of oligomannose type N-glycans and α2,6-linked sialic acid increases during progression from primary to metastatic melanoma. Such changes in glycosylation accompanying specific disease states underscore the importance of analyzing different types of PTMs in the context of the proteome of melanoma-derived EVs and their impact on the mechanism of protein loading into EVs, as well as identifying specific proteins with abnormal PTMs to discover new biomarkers of melanoma.

Tumor-induced modifications in glycan biosynthesis led to the appearance of tumor-associated carbohydrate antigens (TACAs) on the surface of tumor cells, which play a key role in tumor progression and its spread [34–36]. The presence of TACAs affects various aspects of carcinogenesis, such as cell-to-cell and cell-to-extracellular matrix (ECM) interactions, tumor cell survival and proliferation, tumor invasion, metastasis, angiogenesis, chronic inflammation, immune evasion and avoidance of apoptosis. TACAs have also the potential to be used as powerful diagnostic and prognostic tools in clinical practice, including those present in the EV proteome [37]. One of the most commonly observed modifications concerning TACAs is higher expression of β1,6-branched N-linked glycans, resulting from increased transcription and/or activity of β1,6-N-acetylglucosaminyltransferase V (GnT-V, EC 2.4.2.155), encoded by the *MGAT5* gene. It has been shown that β1,6-branched N-glycans are not synthesized by melanocytes or by early melanoma *in situ* cells, although they are often found in fully developed melanoma *in situ* as well as in invasive and metastatic melanoma [38]. Moreover, the presence of β1,6-branched N-glycans is functionally associated with decreased cell adhesion to the ECM, increased cell motility, vascular reconstitution and enhanced metastatic potential, as demonstrated *in vitro* and in mouse models [38,39]. On the other hand, another enzyme, N-acetylglucosaminyltransferase III (GnT-III, EC 2.4.1.144), encoded by the *MGAT3* gene, catalyzing the addition of the so called bisecting N-acetylglucosamine (GlcNAc) in N-glycans, is regarded to be a cancer inhibiting factor by diminishing migratory and invasive properties of cancer [40]. The incorporation of a bisecting GlcNAc residue induces a conformational change in the glycan, impeding further processing and biosynthesis of β1,6-branched N-glycans catalyzed by GnT-V [41].

Although the role of GnT-V and GnT-III and their relationship have been extensively studied in melanoma over the years, there is no information on whether and how the catalytic activity of these enzymes affect the cargo of EVs released by melanoma cells. Therefore, the aim of this liquid chromatography-tandem mass spectrometry (LC-MS/MS)-based study was to investigate whether changes in the glycosylation profile of melanoma cells induced by overexpression of GnT-III and GnT-V would result in qualitative and quantitative changes in the proteome composition of EXOs and MVs released by them. In addition, the selective repertoires of proteins found in the EV cargo were discussed, with a particular focus on enriching a distinct set of proteins based on the glycosylation status of melanoma cells. This exploration delves into the functional implications in melanoma and the clinical relevance of such protein profiles.

## 2. Materials and methods

### 2.1. Materials

Lumi-Light PLUS Western Blotting Kit (including anti-mouse and anti-rabbit IgG-HRP secondary antibodies), penicillin/streptomycin solution, protease and phosphatase inhibitor cocktails, trypsin-EDTA solution, SpeedBeads™ GE45152105050250 and GE65152105050250 were purchased from Merck Group (Darmstadt, Germany). RPMI 1640 GlutaMAX™-I medium, fetal bovine serum (FBS), Alamar Blue cell viability reagent, 0.4 % trypan blue solution, SYBR™ Green Universal Master Mix, MicroBCA Protein Assay kit, Acclaim PepMap trap (100C18, 75 μm × 20 mm, 3 μm particle, 100 Å pore size) and analytical (Acclaim PepMap RSLC C18, 75 μm × 500 mm, 2 μm particle, 100 Å pore size) columns were obtained from Thermo Fisher Scientific (Waltham, MA, USA). The 2x concentrated Laemmli sample buffer was obtained from Bio-Rad Laboratories (Hercules, CA, USA). Biotinylated lectins: *Phaseolus vulgaris* leukoagglutinin (PHA-L), *Phaseolus vulgaris* erythroagglutinin (PHA-E), *Sambucus nigra* agglutinin (SNA), and *Maackia amurensis* agglutinin (MAA) were purchased from Vector Laboratories Inc. (Burlingame, California, United States). Polyvinyidene fluoride membranes (PVDF) were from Millipore (Burlington, MA, USA). Total RNA Midi Kit and Tran-Scriba Kit were purchased from A&A Biotechnology (Gdańsk, Poland). Trypsin/Lys-C Mix was the product of Promega (Madison, WI, USA). High-performance liquid chromatography (HPLC) solvents were of LC/MS grade and were purchased from JT Baker (Phillipsburg, NJ, USA). All remaining chemicals were of analytical grade, commercially available.

### 2.2. Cell culture for isolation of EXOs and MVs

Three variants of the WM266-4 metastatic cutaneous melanoma cell line [42] were used in this study as the experimental model: WM266-4$^{GnT-V}$ cells that stably overexpressed the *MGAT5* gene, WM266-4$^{GnT-III}$ cells that stably overexpressed the *MGAT3* gene and WM266-4$^{pIRESneo}$ mock control cells [43]. The cells were cultured and propagated in RPMI-1640 medium with GlutaMAX™–I supplemented

with 10 % fetal bovine serum, standard antibiotics (100 U/mL penicillin and 100 μg/mL streptomycin) and the selection agent (100 mg/ml G418 sulfate) with 5 % $CO_2$ and 37 °C. Cells were passaged after reaching approximately 85 % confluence. Before collecting cells and conditioned media for further experiments, the cells were cultured for 24 h in serum-free medium. All cell cultures were regularly tested for mycoplasma using standard tests (MycoAlert® PLUS Mycoplasma Detection Kit, Lonza).

### 2.3. Isolation of EXOs and MVs

The procedure for isolation of EXOs and MVs was as described in detail in Refs. [30,31,44]. Briefly, cells and cellular debris were discarded from conditioned media (approx. 200 mL) by sequential centrifugation at 400×*g* (5 min, 4 °C) and 4500×*g* (20 min, 4 °C), followed by concentration of the supernatant using a low-vacuum filtration (LVF) procedure. To remove larger vesicles, the resulting concentrated media (approx. 2 mL) were centrifuged at 7000×*g* (20 min, 4 °C). To obtain MV pellets, additional centrifugation was performed at 18 000×*g* (20 min, 4 °C). Finally, the collected supernatants were ultracentrifuged at 150 000×*g* (90 min, 4 °C) to obtain EXO pellets.

### 2.4. Verification of the purity of the obtained samples of EXOs and MVs

Transmission electron microscopy (TEM) and nanoparticle tracking analysis (NTA) were performed to verify the purity of the obtained samples of EXOs and MVs as described in Refs. [31,33], respectively. Detection of specific EXO and MV markers by Western blotting was performed as described in Ref. [33]. For this purpose, the following primary antibodies were used: anti-heat shock protein 70 (HSP70) (1:2000, clone C92F3A-5, Santa Cruz Biotechnology, Dallas, TX, USA, cat. no. sc-66048), anti-ADP ribosylation Factor 6 (ARF6) (1:500, clone 3A-1, Santa Cruz Biotechnology, cat. no. sc-7971), anti-CD9 (1:2000, clone 4A2, Santa Cruz Biotechnology, cat. SAB1402143), anti-CD63 (1:2000, clone RFAC4, Santa Cruz Biotechnology, cat. CBL553), and mouse monoclonal anti-β-actin (1:10 000, clone AT-15, Sigma-Aldrich, St. Louis, MI, USA). Anti-mouse IgG-HRP (1:400) was used as a secondary antibody. EV markers were detected using chemiluminescent substrates for HRP and ChemiDoc imaging system (Bio-Rad Laboratories).

### 2.5. Analysis of glycosylation by flow cytometry

Analysis of the glycosylation of the surface of WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells as well as EVs derived by them was performed by flow cytometry using selected lectins (Table 1) as described in Ref. [45] with minor modification [33]. Briefly, cells ($7 \times 10^4$) as well as EXOs and MVs (isolated from 10 mL of conditioned media per sample) were incubated for 45 min on ice with biotinylated lectins (20 μg/mL PBS). The cells and EVs then were incubated with fluorescein isothiocyanate (FITC)-extravidin conjugate (25 μL/mL) under the same conditions after washing with PBS and washed again.

**Table 1**
Sugar-binding specificities of lectins used for flow cytometric studies.

| Lectin | Recognized lectin-specific glycan structures | Abbreviation used |
|---|---|---|
| *Maackia amurensis* agglutinin | NeuAcα2,3Gal- | MAA |
| *Phaseolus vulgaris* erythroagglutinin | Galβ1,4GlcNAcβ1,2Man (the bisecting GlcNAcβ1,4Man is essential) | PHA-E |
| *Phaseolus vulgaris* leucoagglutinin | Galβ1,4GlcNAcβ1,6Manα-Galβ1,4GlcNAcβ1,2Manα-(β1,6-branched structures of N-glycans) | PHA-L |
| *Sambucus nigra* agglutin | NeuAcα2,6Gal- | SNA |

Cells were assessed for fluorescence in Aurora CS spectral cell sorter (Cytek, California, USA). For each sample, a total of $10^4$ cells were analyzed. EXOs and MVs were assessed for fluorescence also in Aurora CS spectral cell sorter. For each sample, a total event count of $10^6$ EVs was analyzed with 0.1 % of FSC value. Statistical analysis of the data was carried out using OriginPro 2024 software.

### 2.6. Analysis of the proteome composition of EXOs and MVs by tandem mass spectrometry coupled with liquid chromatography (LC-MS/MS)

The analysis of EXOs and MVs released by WM266-4 cells in the three genetic variants was done in four replicates per experimental group. EXOs and MVs lysis was performed as described in Surman et al. [28] with minor changes. Samples were sonicated in the Bioruptor Pico (Diagenode) for about 7 min using 30s/30s ON/OFF cycle. The pellets obtained after protein precipitation were firstly solubilized in 5 μL of 1 % SDS, 0.1 M Tris, pH 7.5 and then diluted up to 330 μL with 50 mM HEPES, pH 8. The resulting samples were further processed for LC-MS/MS analysis as described in Surman et al. [28], however, following the updated SP3 protocol [46]. The LC-MS/MS analysis was performed using the Q Exactive mass spectrometer coupled with the UltiMate 3000 RSLCnano System (both Thermo Fisher Scientific). Peptides were loaded onto a trap column in 2 % acetonitrile with 0.05 % trifluoroacetic acid (TFA) at a flow rate of 5 μL/min and separated on an analytical column with a 240 min gradient from 2 % to 40 % acetonitrile in 0.05 % formic acid at a flow rate of 250 nL/min. The Q Exactive was operated in a data-dependent mode using Top12 method. The full MS and MS/MS spectra were acquired at resolutions of 70 000 and 17 500, respectively. The performance of the LC-MS/MS system was monitored using the QCloud quality control system [47].

#### 2.6.1. Identification and quantification of proteins in EXO and MV samples

The collected LC-MS/MS data were processed with MaxQuant (version 2.1.4.0) [48] and searched using an integrated Andromeda search engine [49] against the UniProtKB database limited to *Homo sapiens* taxonomy (20 424 sequences; downloaded on July 31, 2023). The false discovery rate (FDR) for the peptide and protein identification was set to 1 %. Label free quantification (LFQ) as well as intensity-based absolute quantification (iBAQ) [50] were carried out. Data acquired for EXO and MV samples were processed separately.

#### 2.6.2. Quantitative proteomic analysis

Perseus platform (version 2.0.10.0) was used in the next step of analysis. All contaminants, the proteins from the decoy database, and proteins only identified by site were excluded from the study. Quantitative analysis of proteins identified in EXOs was performed using log-transformed LFQ intensities. Data matrix was filtered for proteins with at least three valid values in at least one group and then missing values were imputed. The Student's t-test followed by the permutation-based FDR correction set to 0.05 was performed to reveal changes in the protein abundances between EXOs derived from three genetic variants of WM266-4 cells. Only proteins with at least three valid LFQ intensity values in at least one of two tested groups were considered for statistical analysis. All differential proteins were identified based on at least 2 peptides. Proteins with a fold change of at least 1.5 were considered as differential. Entire protein lists and complete quantitative/statistical analysis were provided in Supplementary Data 2. LFQ analysis could not be used to compare quantity of proteins in MVs due to big difference between number of identified proteins in each MV populations. Thus, iBAQ values were used to characterize different MV groups. Data were processed in Perseus platform as in the case of LFQ intensity-based analysis performed for EXOs. Data were log-transformed and means of the iBAQ values were calculated for proteins with at least 3 valid iBAQ values in a given group to assess their abundance within MVs released by WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells.

*2.6.3. Bioinformatic analysis*

For further analysis of obtained proteomic data several diverse bioinformatics tools were used (Fig. 1). For the functional analysis, the final protein lists were created manually and included only proteins identified in three or four biological replicates of EXO or MV samples and with at least two peptides. FunRich 3.1.4 software (http://funrich.org/index.html) [51], with UniProt database (release November 2023) as a reference, was used to create Venn diagrams and analyze gene ontology (GO) for cellular compartment, molecular function and biological processes. For each GO term, the six categories with the highest statistical significance of protein enrichment within the corresponding category (calculated as −log10(p-value)) are presented as graphs. Interactomes were prepared using string v11.0 (https://string-db.org, accessed on September 10, 2024).

Data on glycosylation status of selected vesicular proteins were obtained from GlyGen (https://glygen.org) [52] and GlyConnect (https://glyconnect.expasy.org) [53] databases. The GlyGen database was used to show the number of reported glycosylation sites, reported sites of N-glycosylation, reported sites with N-glycans, and the number of Asn in the protein chain on which N-glycans are present. The GlyConnect database was used to show the types of identified N-glycans on individual Asn. The results obtained using glycosylation-related databases were shown in tables.

The prognostic values of proteins identified in EXOs and MVs were analyzed using data presented in The Human Protein Atlas database. RNA expression values in skin cancer cell lines were presented as transcripts per million (nTPM), RNA expression in tissue from The Cancer Genome Atlas (TCGA) as fragments per kilobase per million mapped fragments (FPKM). Patients were divided into two groups, under cut-off value or over cut-off value, and the relationship between survival and FPKM was examined. The cut-off values refer to the FPKM value that yields the maximum survival difference between the two groups with the lowest log-rank P-value. The correlation between mRNA expression levels and patient survival was used to obtain P-value for the Kaplan-Meier plot [54–56]. The results obtained were shown in tables.

To analyze the involvement of proteins present in the respective EV populations in melanoma pathways, KEGG pathways and DISEASE pathways were searched using string-db.org. Terms strictly assigned to melanoma were used: hsa05218 in the KEGG database (72 proteins) and DOID:1909 in the DISEASES database (46 proteins). The proteins identified in EVs were then compared separately to the list of proteins in both pathways. The KEGG and DISEASES melanoma pathways were presented as string graphs. The string graphs were prepared in a full STRING network, the edges of the network symbolized the strength of the data support, and the minimum interaction score was set as the highest confidence level (0,900). The results obtained were presented in table.

The DisGeNET (https://disgenet.com/) database was used to analyze proteins pre-sent in melanoma-derived EVs in terms of the number of diseases associated with the genes, the gene disease association (GDA) score, the diseases specificity index (DSI) for the gene and the diseases pleiotropy index (DPI) for the gene. For analysis, 'melanoma' was used as a disease category, with Concept Unique Identifier (CUI): C0025202, 4792 genes, and 623 snaps. The GDA score is computed as SGDA= C + M + I + L + T, where each letter corresponds to a different type of sources representing GDA. Namely, C represents the number of CURATED sources, M - the number of MODEL sources, I - the number of INFERRED sources, L - the number of publications, and T - the number of Clinical Trials. DSI shows the association of a gene/gene variant with multiple diseases and represents whether the gene is involved in one or

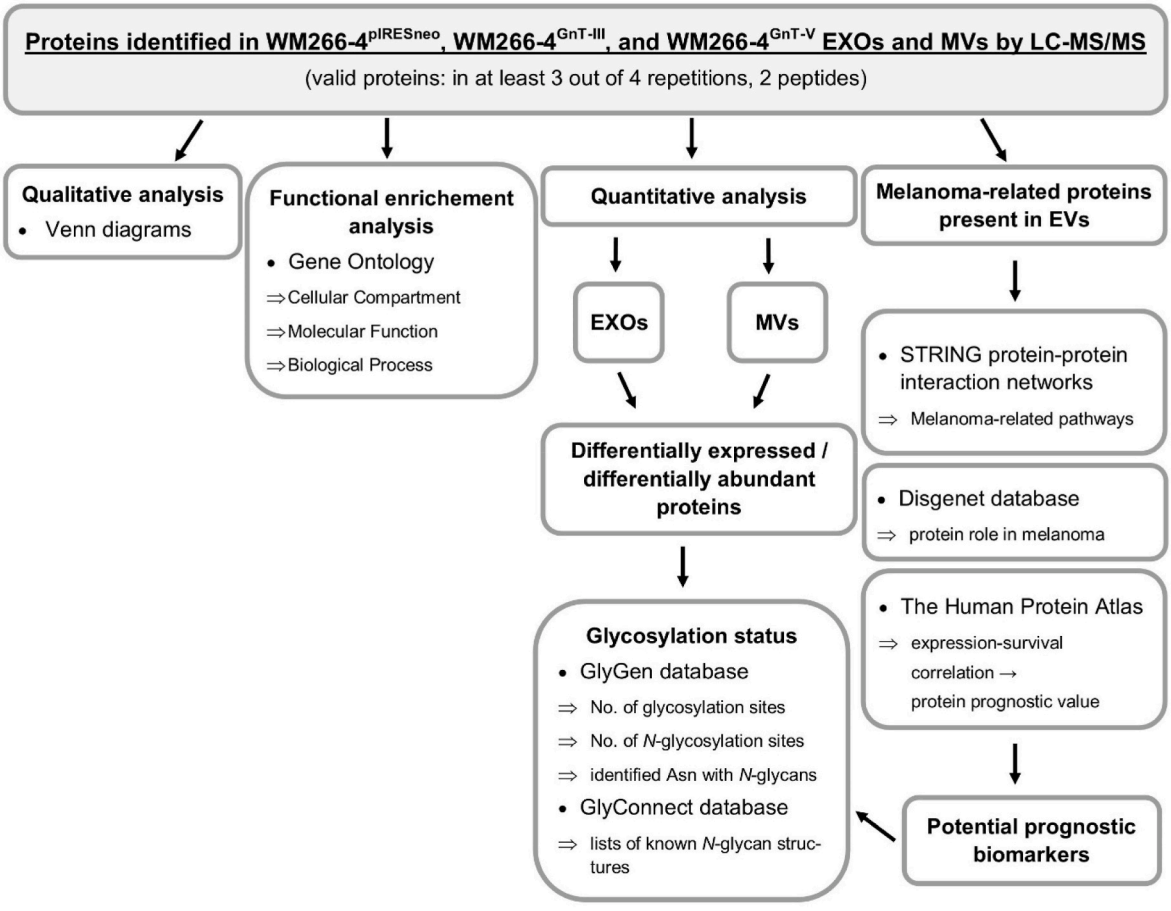



**Fig. 1.** Workflow scheme summarizing different stages of bioinformatic analysis.

multiple diseases. DSI values range from 0.23 to 1, where 1 means the gene is associated only with one disease. DPI takes into account whether the multiple diseases associated with a gene/gene variant belong to the same Medical Subject Headings (MeSH) disease class (*e.g.* neoplastic disease) or belong to completely different disease classes. DPI values range from 0 to 1, with larger DPI values symbolizing a larger and more diverse number of MeSH categories within a gene [57]. The proteins present in the respective EV populations were compared with proteins encoded by genes present in the melanoma category. Only proteins with DSIs in range from 1 to 0.40 were presented in table. In addition, proteins were presented in the results only if possible, a melanoma-associated function could be found in the literature. The functions of individual proteins were also shown in table.

## 3. Results

### 3.1. Characterization of the isolated samples of EVs

Characterization of EXOs and MVs released by the three genetic variants of WM266-4 cells was performed using TEM, NTA and WB to meet MISEV2023 criteria. As shown by TEM, there were no intact cells, cell organelles or other cellular debris in the EV samples (Fig. 2A). WB analysis (Fig. 2B) of the expression of selected exosomal and microvesicular protein markers showed the presence of both an exosomal marker (HSP70, CD9, and CD63) and a microvesicular marker (ARF6) in the isolated samples.

Analysis of exosomal and microvesicular samples by NTA (Fig. 3) showed that the EXO and MV populations were heterogenous in size, based on the number of peaks on each graph. EXOs had an average diameter between 56 and 104 nm, while MVs were observed as objects ranging in size from 130 to 405 nm.

### 3.2. Effect of MGAT3 and MGAT5 overexpression in WM266-4 cells on cell and extracellular vesicle surface glycosylation

Overexpression of *MGAT3* gene in WM266-4 cells (WM266-4$^{GnT-III}$ cells) resulted in a 25 %, 30 %, 30 % and 35 % decrease in the number of positive cells stained with PHA-L, MAA, SNA and PHA-E lectins, respectively (Fig. 4A). Nevertheless, for PHA-E and PHA-L staining, an increase in relative fluorescence intensity (RFI) values of 2.25 and 1.3 times was observed (Fig. 4B). Induction of *MGAT5* gene overexpression in WM266-4 cells (WM266-4$^{GnT-V}$ cells) resulted in a 15 % decrease in the number of positive cells stained with the two lectins (PHA-E and MAA), a 1.2-fold increase after PHA-L staining, but had no effect on the number of SNA-positive cells (Fig. 4A). As for RFI values, they were found to increase with PHA-E and MAA staining, by 1.35 and 1.5 times, respectively (Fig. 4B).

Overexpression of *MGAT3* and *MGAT5* genes resulted in a 1.8-fold increase in the percentage of MVs released by both genetic variants (WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells) showing N-glycans with bisecting GlcNAc (positive reaction with PHA-E), while in EXOs the increase was 1.2-fold and 1.4-fold for these lines, respectively, in comparison to WM266-4$^{pIRESneo}$ cells (Fig. 4CE). There was also an increase in the percentage of EVs showing the presence of β1,6-branched N-glycans on their surface (positive reaction with PHA-L), but this effect was more pronounced in EXOs (2.3-fold for those released by WM266-4$^{GnT-III}$ cells and 1.9-fold for those released by WM266-4$^{GnT-V}$ cells) than MVs (1.2-fold for those released by WM266-4$^{GnT-III}$ cells and 1.6-fold for those released by WM266-4$^{GnT-V}$ cells) (Fig. 4CE). The effect of cell transfection on the percentage of EVs showing α2,6-linked NeuNAc (positive reaction with SNA) was similar in EXOs and MVs (1.4–1.5-fold), except for the weaker increase found in MVs released by WM266-4$^{GnT-III}$ cells (Fig. 4CE). For α2,3-bound NeuNAc (positive reaction with MAA), an increase in the percentage of EVs was observed in the 1.4–1.5-

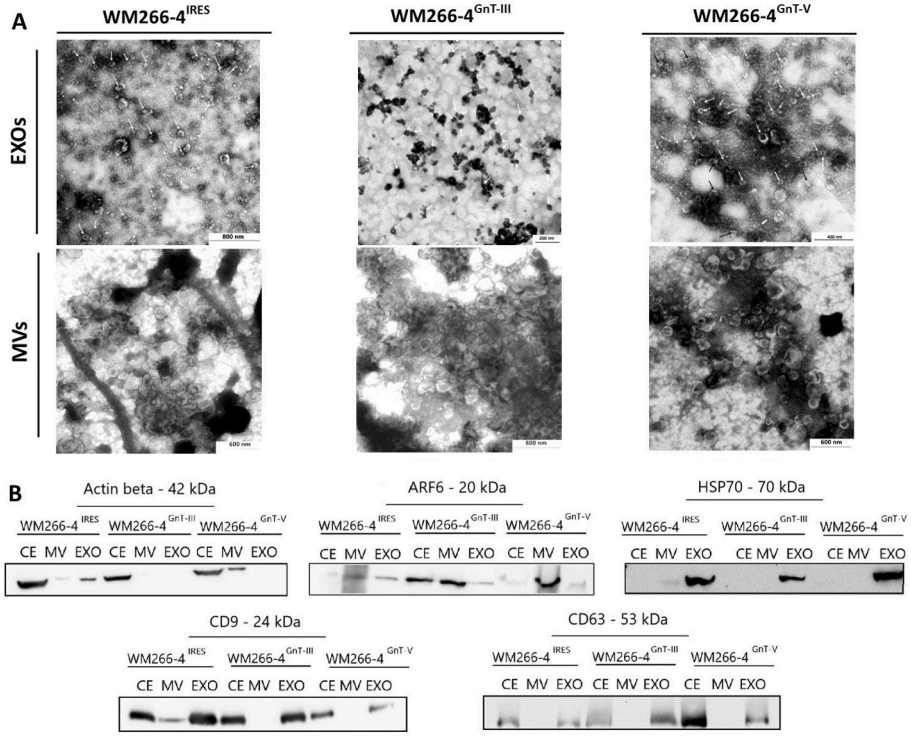



**Fig. 2.** Characterization of EV samples isolated from conditioned media of WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ melanoma cells. A. Transmission electron microscopy (TEM) image of EXOs and MVs derived from the genetic variants of WM266-4 cells. B. Representative Western blot of exosomal and microvesicular markers in whole-cell protein extracts (CE), exosomes (EXO), and microvesicles (MV) samples. Samples containing 50 μg of protein were separated by 10 % polyacrylamide gel electrophoresis (SDS-PAGE) and transferred into PVDF membrane, and then probed with anti-HSP70 (1:2000 dilution), anti-ARF6 (1:500), anti-CD9 (1:2000), anti-CD63 (1:2000) and anti-β-actin (1:10 000) as primary antibodies and anti-mouse IgG-HRP (1:400) as a secondary antibody.

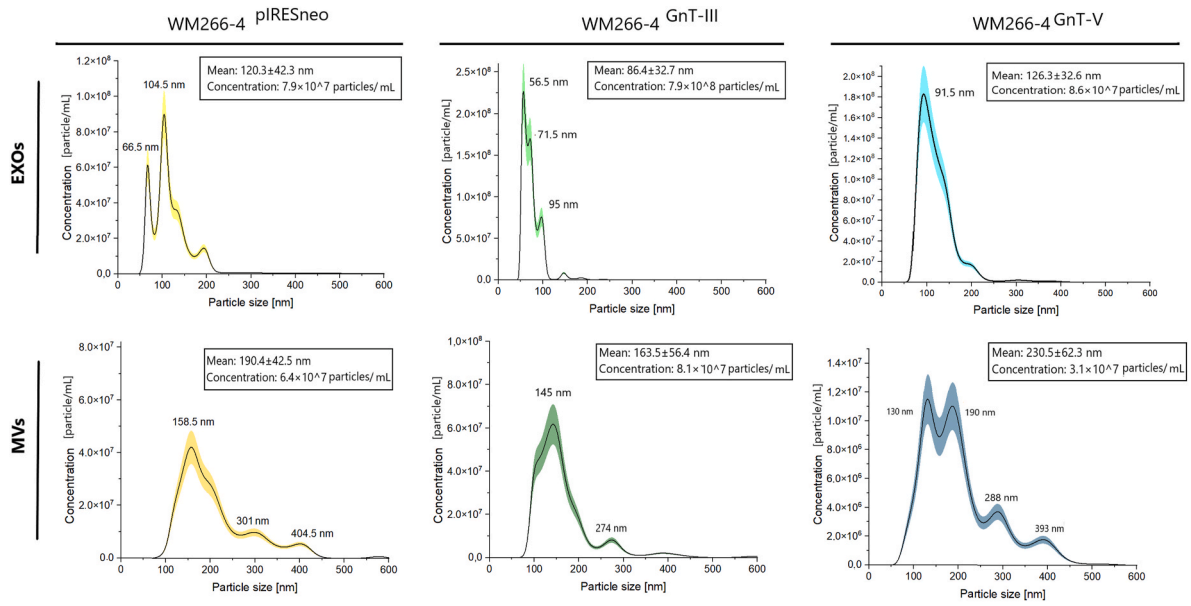



**Fig. 3.** Nanoparticle tracking analysis (NTA) of EXO and MV samples released by WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ melanoma cells. The average results from five independent measurements for each of the genetic variants are shown in the graphs. The colored area represents the standard deviation.

fold range, but for MVs released by WM266-4$^{GnT-V}$ cells, this parameter decreased by 25 % (Fig. 4CE). Additionally, the RFI value for PHA-E staining increased 4.9-fold and 3-fold only in MVs released by WM266-4$^{GnT-III}$ cells and WM266-4$^{GnT-V}$ cells, respectively (Fig. 4DF). With PHA-L staining, there was a 10–50 % decrease in RFI in both EXOs and MVs (Fig. 4DF). Increases in RFI value relating to MAA and SNA staining were found in EXOs and MVs, ranging from 1.4-fold to 7.5-fold, but no changes were seen in MVs (MAA staining) and EXOs (SNA staining), both released by WM266-4$^{GnT-III}$ cells (Fig. 4DF). This confirms that *MGAT3* and *MGAT5* overexpression in WM266-4 melanoma cells had a visible impact on surface glycosylation in EVs released by them.

### 3.3. Proteins identified in MVs and EXOs released by WM-266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, WM-266-4$^{GnT-V}$ cells

#### 3.3.1. Qualitative proteomic analysis

Comprehensive proteomic profiling of MVs and EXOs released by WM-266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM-266-4$^{GnT-V}$ cells was performed using gel-free shotgun nanoLC–MS/MS. A total number of 1770 microvesicular and 704 exosomal proteins were identified by at least two peptides in all biological and technical replicates for EVs isolated from WM266-4$^{pIRESneo}$ cells (Fig. 5A). For MVs and EXOs derived from WM266-4$^{GnT-III}$ cells, 750 and 541 proteins were identified, respectively. MVs and EXOs derived from WM266-4$^{GnT-V}$ cells carried 1323 and 853 proteins, respectively. In addition, 597 (~32 %), 454 (~54 %) and 699 (~47 %) proteins were common to both EV types derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, respectively. Furthermore, comparative analyses based on Venn diagram of MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ cells showed that 488, 5, and 43 proteins were unique to the respective MV populations. Similar analyses for EXOs showed that 45, 28, and 168 identified proteins were present only in WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ EXOs, respectively (Fig. 5B).

In addition, the lists of identified microvesicular and exosomal proteins were subjected to a separate Gene Ontology (GO) analysis of the three main categories, *i.e.*, cellular compartment, molecular function, and biological process, using the UniProt database as a reference. For each analysis, the six categories with the highest enrichment were presented as graphs, and the full reports were provided in Supplementary Data 2. In terms of cellular origin, 64.2 % of WM266-4$^{pIRESneo}$, 62.9 % of WM266-4$^{GnT-V}$ and 72.5 % of WM266-4$^{GnT-III}$ exosomal proteins were assigned to the 'Extracellular exosome' category (Fig. 6A). The second most enriched GO category for exosomal proteins was 'Cytosol', with an average of 53 % of proteins. An analogous analysis for MVs derived from all three analyzed cell lines showed that approximately 60 % of their microvesicular proteins were related to 'Cytosol' (Fig. 6B). Other highly enriched categories were related to cytoplasmatic proteins (~42 % of identified proteins), membrane proteins (~33 % of identified proteins) and focal adhesion-related proteins (~15 % of identified proteins). Finally, the enrichment of both MVs and EXOs in cytoplasm- and cell membrane-associated proteins is closely related to their biogenesis.

Furthermore, GO analysis in the "molecular function" category showed that EXOs derived from WM-266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM-266-4$^{GnT-V}$ cells were generally enriched in proteins involved in binding other proteins (such as cadherins or integrins) or nucleic acids (Fig. 7A). EXOs were also enriched in proteins associated with threonine-type endopeptidase activity proteins accounting for about 2 % of all identified exosomal proteins. As with exosomal proteins, the vast majority of proteins present in MVs were associated with protein and RNA binding (Fig. 7B). On the other hand, in contrast to EXOs, MVs contain a significant percentage of proteins involved in nucleotide binding or hydrolysis. This confirms the role of the protein cargo of MVs in regulating cellular signaling, metabolic processes and RNA synthesis, in which transferred nucleotides actively participate.

GO analysis for involvement in biological processes revealed significant differences in the functions performed by EXOs and MVs (Fig. 8). The greatest similarity between both types of vesicles was the enrichment in proteins associated with neutrophil degranulation and regulation of mRNA stability. As for differences, about 12 % of the exosomal proteins identified in all WM266-4 cell variants were involved in post-translational protein modification, 5.5 % in the Wnt signaling pathway and the planer cell polarity pathway, and about 5 % of them were correlated with the regulation of cellular amino acid metabolic processes or antigen processing and MHC-I-mediated presentation of exogenous peptide antigen in a TAP-dependent manner. MVs proteins were shown to be involved in the catabolism of nuclear-transcribed mRNA (nonsense-mediated decay) and SRP-dependent co-translational targeting of proteins to the membrane, approximately 4.5 % in both categories.

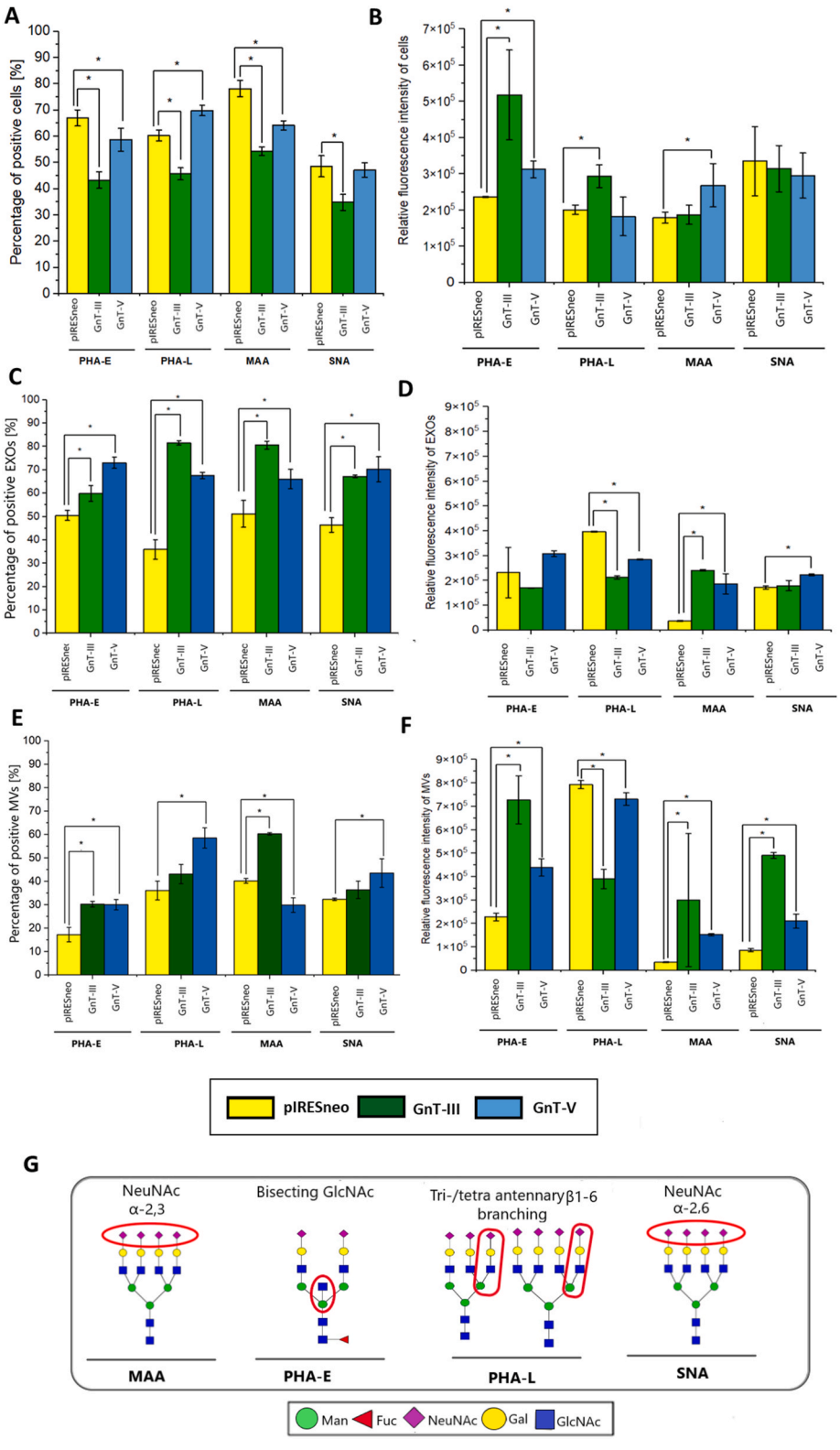



**Fig. 4.** Flow cytometry analysis of surface glycosylation of WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ cells as well as EXOs and MVs released by them. A, C, E. Diagrams showing the mean percentage of positive cells, EXOs and MVs, respectively, in each lectin staining. B, D, F. Diagram showing the mean relative fluorescence intensity (RFI) in each lectin staining for cells, EXOs and MVs, respectively. G. Specificity of lectins used in flow cytometric analysis. All diagrams present means from three independent experiments ± SD. Statistically significant differences are marked as *. Respective histograms for each staining are provided in Supplementary Data 1.

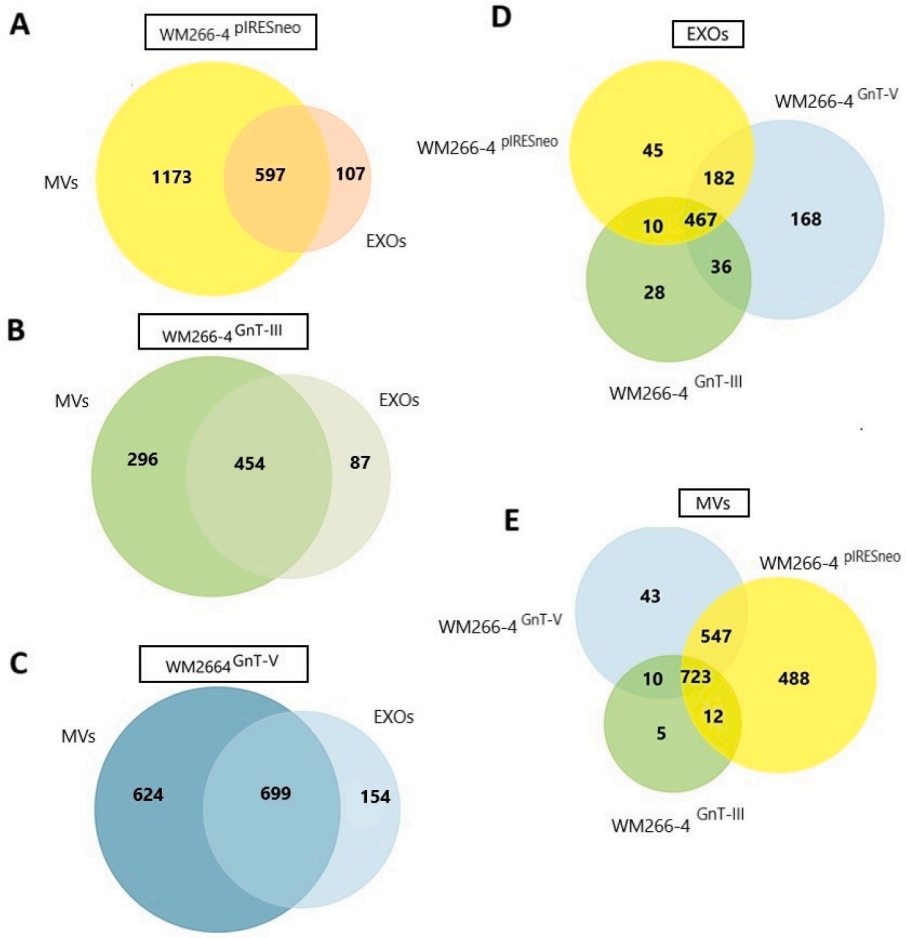



**Fig. 5.** Basic comparative analysis of proteins identified by nanoLC-MS/MS in EXOs and MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$, and WM266-4$^{GnT-V}$ cells. (A) Venn diagrams comparing numbers of proteins identified in EXOs and MVs derived from the same genetic variant of WM266-4 cells, that is, WM266-4$^{pIRESneo}$ cells (A), WM266-4$^{GnT-III}$ cells (B), and WM266-4$^{GnT-V}$ cells (C). Venn diagrams comparing proteins between EXOs (D) and MVs (E) derived from three genetic variants of WM266-4 cells.

Next, protein cargoes of both EXOs and MVs were screened for the presence of proteins closely related to the N-glycosylation process (GO: 0006487, protein N-linked glycosylation) (Table 2). First, EVs were checked for proteins whose expression should be increased in WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, *i.e.*, GnT-III and GnT-V, respectively. GnT-V was present in both EXOs and MVs derived from WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, as well as in WM266-4$^{pIRESneo}$ cells, while the presence of GnT-III was not found in any sample. This may be related to the fact that melanoma cells with *MGAT5* overexpression show higher malignant potential. Therefore, melanoma-derived EVs may transfer GnT-V to less malignant subclones of melanoma cells in tumors. In addition, α-1,3-mannosyl-glycoprotein 4-β-N-acetylglucosaminyl-transferase B (MGAT4B), which is required for the synthesis of tri- and tetra-antennary N-glycans, was present only in EXOs derived from WM266-4$^{GnT-V}$ and WM266-4$^{pIRESneo}$ cells. Furthermore, α-1,3-mannosyl-glycoprotein 2-β-N-acetylglucosaminyltransferase (MGAT1) required for the synthesis of hybrid and complex type N-glycans was identified in EXOs derived from WM266-4$^{pIRESneo}$ cells and in both EV populations derived from WM266-4$^{GnT-V}$ cells. Finally, ST3 β-galactoside α-2,3-sialyltransferase 4 (ST3GAL4), responsible for the terminal addition of α-2,3-linked NeuNAc in glycoproteins, was identified in all samples except MVs derived from WM266-4$^{GnT-V}$ cells and EXOs derived from WM266-4$^{GnT-III}$ cells. Increased expression of α2,3-linked sialic acids on the cell surface has been found to promote the interaction of melanoma cells with fibronectin [58]. Moreover, manipulation of miR-1180–3p expression identified the boosting effect of ST3GAL4 on melanoma cell migration [59].

GO analysis was also used to select EV proteins which may contribute to the transfer of metastatic potential to recipient cells (Supplementary Table S1). Since changes in cell glycosylation may affect the sorting of protein into EVs, the presence of these proteins was compared between EXOs and MVs derived from all three genetic variants of WM266-4 cells and presented in Supplementary Table S1. Most of the selected proteins were present in both EXOs and MVs derived from all variants of WM266-4 cells. On the other hand, proteins such as vascular endothelial growth factor receptor 1 (VEGFR) or matrix metalloproteinase 14 (MMP14) were identified only in EVs derived from WM266-4$^{pIRESneo}$ cells. Finally, proteins such as disintegrin and metalloproteinase domain-containing protein 10 (ADAM10), integrin β1, or cathepsin D (Cat D) were present only in EXOs derived from WM266-4$^{pIRESneo}$ cells, but not in MVs. However, they were identified in both EXOs and MVs derived from WM266-4 cells with induced overexpression of *MGATs*, i.e., WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells.

Autophagy is a pathway for the degradation of long-lived cytoplasmic proteins, protein complexes or damaged organelles [60]. Moreover, autophagy and exosome biogenesis are interconnected at the stage of multivesicular bodies, which fuse either with the cell membrane, resulting in the release of exosomes, or with the autophagosome, forming amphisomes degraded by autophagy [61,62]. The diversity of proteins involved in autophagy was identified during GO analysis in EXOs released from WM266-4 cells overexpressing *MGAT3* and *MGAT5* genes (Supplementary Table S2). HSP90, present in EXOs released by all

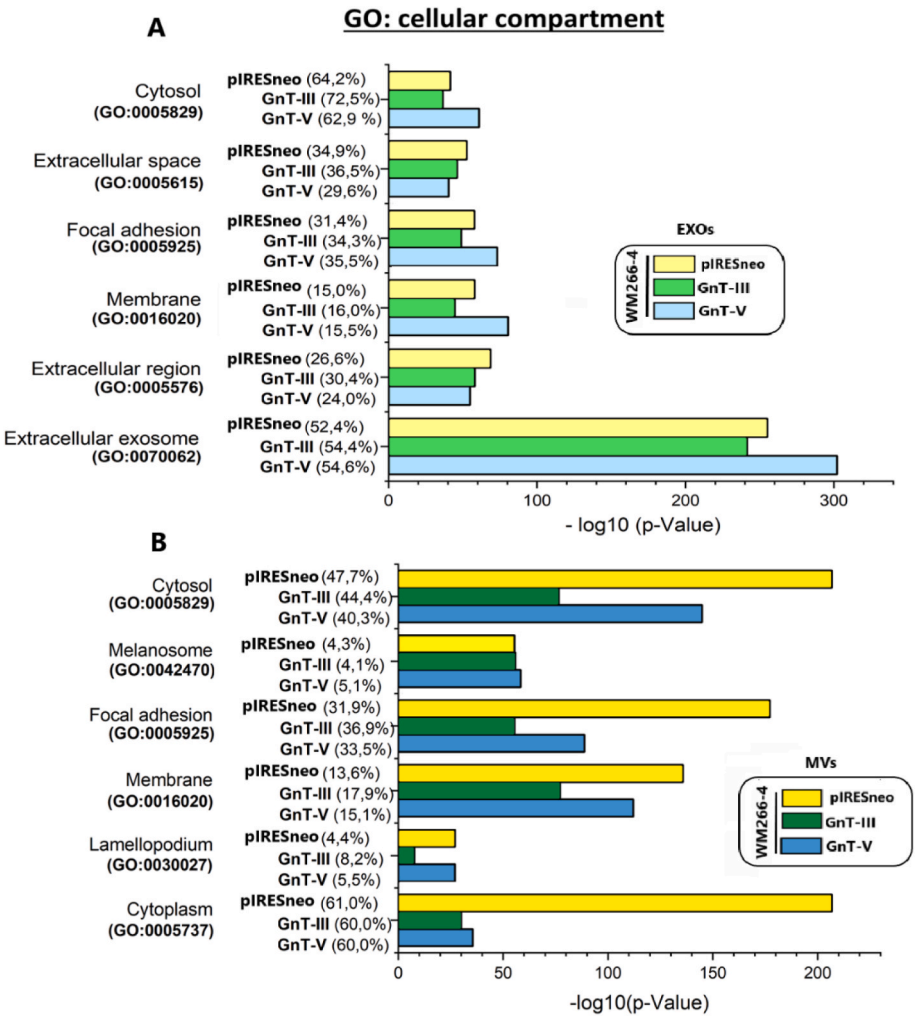



**Fig. 6.** Gene ontology (GO) analysis of proteins identified in EXOs (A) and MVs (B) derived from WM-266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM-266-4$^{GnT-V}$ performed with the use of FunRich 2.0 software with UniProt database as a reference. For the GO term "cellular compartment", six categories with the highest statistical significance of protein enrichment within the specific category ($p < 0.001$) are presented as graphs. The percentages in parentheses represent the percentage of proteins identified for a given GO category out of all proteins identified in a given population of EXOs or MVs. Complete results of GO analysis are provided in Supplementary Data 2.

genetic variants of WM266-4 cells, is not only involved in mitophagy and chaperon-mediated autophagy, but is also a target for anticancer drugs in melanoma (e.g., KU135, Novobiocin, H-15, H-10 or 17-AAG [63]). EVs also carried the tumor microenvironment-associated protein S100A8/A9, which serves as a novel prognostic marker for metastasis and survival of patients with metastatic melanoma and predicts response to pembrolizumab immunotherapy [64]. Binding of S100A8/A9 to melanoma cell adhesion molecule (MCAM) causes melanoma cells to migrate into lung tissue [65]. This shows that exosomes increase melanoma malignancy by affecting changes in autophagy in recipient cells. Therefore, a better understanding of the interplay between EVs and autophagy seems to be crucial for the prognosis and treatment of melanoma patients. Furthermore, EXOs and MVs derived from all genetic variants of WM266-4 cells transported proteins involved in the inhibition of apoptosis and the development of drug resistance. GO results for these categories are presented in the appendices (Supplementary Tables S3 and S4).

EVs are currently being studied to determine their role in the development of disease states, such as cancer, neurodegenerative, autoimmune diseases, or in graft-versus-host disease. However, EVs play a key role in supporting both physiological and pathological processes. Vesicular transport is crucial for the proper functioning of the skin, including melanin transport. Melanocytes, whose neoplastic transformation leads to melanoma, are among the cells in which the result of vesicular transport is evident in processes such as the distribution of melanin in the skin. Even in the EVs released by melanoma cells, a significant number of proteins responsible for these physiological processes is present, suggesting that melanoma cells are still able to retain some of the physiological functions of melanocytes (Supplementary Table S5).

### 3.3.2. *Quantitative proteomic analyses*

For label-free quantitative analysis, separate approaches were used for EXOs and MVs. The LFQ approach was used for EXOs and the iBAQ approach was used for MVs. Due to the large difference between the number of identified proteins in MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, the use of LFQ for MVs was not possible.

*3.3.2.1. LFQ quantitative proteomic analyses of EXO samples.* For LFQ analysis, two comparisons were performed, *i.e.*, proteins identified in WM266-4$^{pIRESneo}$-derived EXOs were compared with proteins identified in WM266-4$^{GnT-III}$-derived EXOs or WM266-4$^{GnT-V}$-derived EXOs (Table 3). Quantitative analysis of exosomal proteins showed that 35 of WM266-4$^{GnT-III}$ proteins were upregulated, whereas 109 proteins were downregulated compared to WM266-4$^{pIRESneo}$ exosomal proteins.

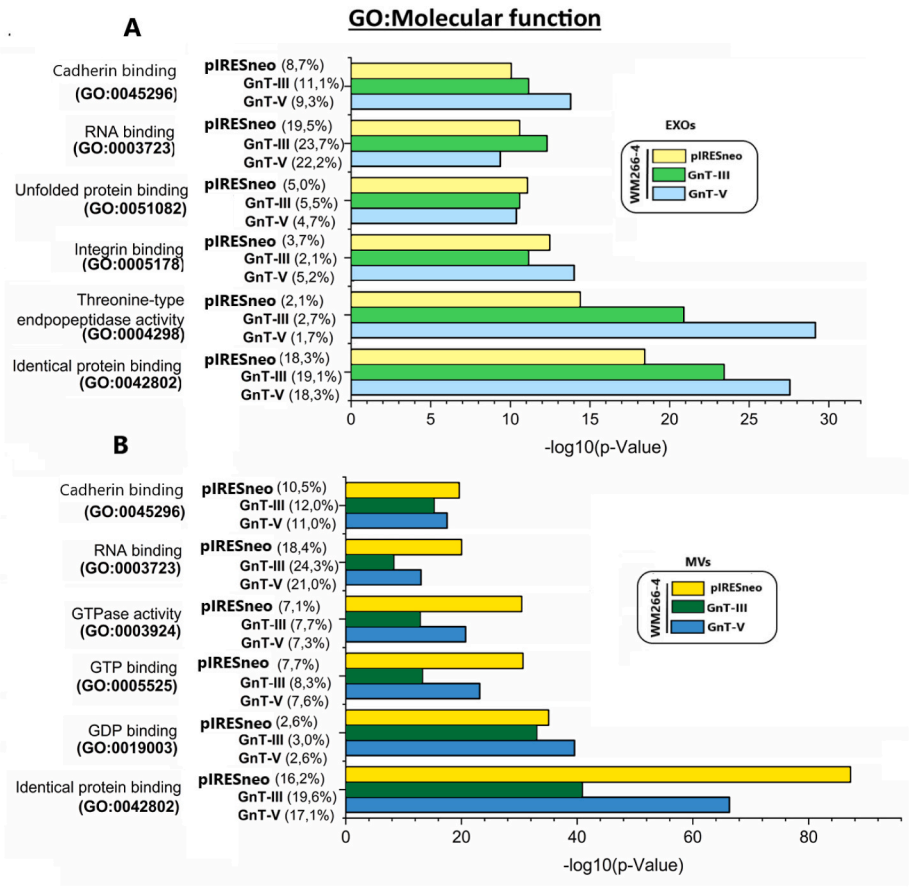



**Fig. 7.** Gene ontology (GO) analysis of proteins identified in EXOs (A) and MVs (B) derived from WM266-4$^{pIRESneo}$, WM2664$^{GnT-III}$, and WM266-4$^{GnT-V}$ performed with the use of FunRich 2.0 software with UniProt database as a reference. For the GO term "molecular function", six categories with the highest statistical significance of protein enrichment within the specific category ($p < 0.001$) are presented as graphs. The percentages in parentheses represent the percentage of proteins identified for a given GO category out of all proteins identified in a given population of EXOs or MVs. Complete results of GO analysis are provided in Supplementary Data 2.

The most highly upregulated protein in WM266-4$^{GnT-III}$-derived EXOs compared to WM266-4$^{pIRESneo}$-derived EXOs is lipocalin-1 (LCN1). LCN1 is a biomarker of various pathologies, including dry eye disease, diabetic retinopathy, chronic obstructive pulmonary disease, and tobacco-related diseases [66–69]. Overexpression of LCN1 has been shown to be observed in malignancies, e.g. in breast cancer it correlates with poor prognosis [70]. Desmocollin-1 (DSC1) and desmoglein 1 (DSG1) were also overexpressed in WM266-4$^{GnT-III}$-derived EXOs. DSC1 is a transmembrane glycoprotein found in desmosomes that plays a key role in cell-to-cell adhesion. DSC1 has previously been implicated in lymph node metastasis in luminal A breast tumors and increased migration and invasion of MCF7 cells *in vitro* [71]. In addition, a temporary reduction of DSG1, a keratinocyte-specific cadherin, played a role in regulating paracrine signaling between melanocytes and keratinocytes. Melanoma cells exploited this intercellular communication by releasing substances that suppressed DSG1 expression in neighboring keratinocytes. Consequently, keratinocytes produced their own paracrine factors that promoted melanoma spread via CXCL1/CXCR2 signaling [72]. Finally, serpin B3, also overexpressed in WM266-4$^{GnT-III}$-derived EXOs, is involved in squamous epithelial differentiation. Upregulated serpin B3 may participate in the apoptotic pathway in epithelial-derived tumor tissues [73]. Serpin B3 was shown to be mutated in melanoma patients who benefited clinically from anticytotoxic T-lymphocyte associated protein 4 (CTLA4) therapy [74].

On the other hand, α-2-macroglobulin (α2M) was identified as a protein overexpressed in WM2664$^{pIRESneo}$ EXOs compared to WM266-4$^{GnT-III}$ EXOs. α2M is expressed in both melanoma cell lines and tissues from melanoma patients. A study of 30 melanoma patients showed that in one patient almost all tumor cells were α2M-positive, whereas in other cases 20 %–50 % of tumor cells were positive [75]. Therefore, it is most likely that the reduced expression of α2M in WM266-4$^{GnT-III}$-derived EXOs is not due to decreased protein expression in melanoma cells but to a N-glycosylation-dependent sorting mechanism. Integrin α2, also downregulated in WM266-4$^{GnT-III}$-derived EXOs, was previously found to be overexpressed in metastatic melanoma cell lines. Moreover, blocking of integrin α2 with anti-integrin α2 antibodies significantly inhibited the migration of metastatic but not primary melanoma cells [76]. In a similar study, integrin blocking with both anti-integrin β1 and anti-integrin α2 antibodies in melanoma cell lines with higher migratory potential resulted in inhibition of cell migration [77]. Therefore, the reason for the reduced migratory capacity of melanoma cells overexpressing *MGAT3* may be correlated with changes in the qualitative and quantitative protein composition in EXOs.

The most up- and down-regulated proteins in WM266-4$^{GnT-III}$-derived EXOs were also searched against glycosylation-related databases (Table 4). In general, the upregulated proteins had fewer N-glycosylation sites than downregulated ones. DSG1 was the only upregulated protein with a known N-glycan structure type, as it has oligomannose structures on Asn10. Two of the downregulated proteins lacked N-glycosylation sites, *i.e.*, tetraspanin-4 and phospho-lipase A1 member A. Other downregulated proteins possessed one to nine N-glycosylation sites, with immunoglobulin superfamily member 3, integrin α2, and α2M possessing the highest number of N-glycosylation sites of all downregulated proteins in WM266-4$^{GnT-III}$-derived EXOs.

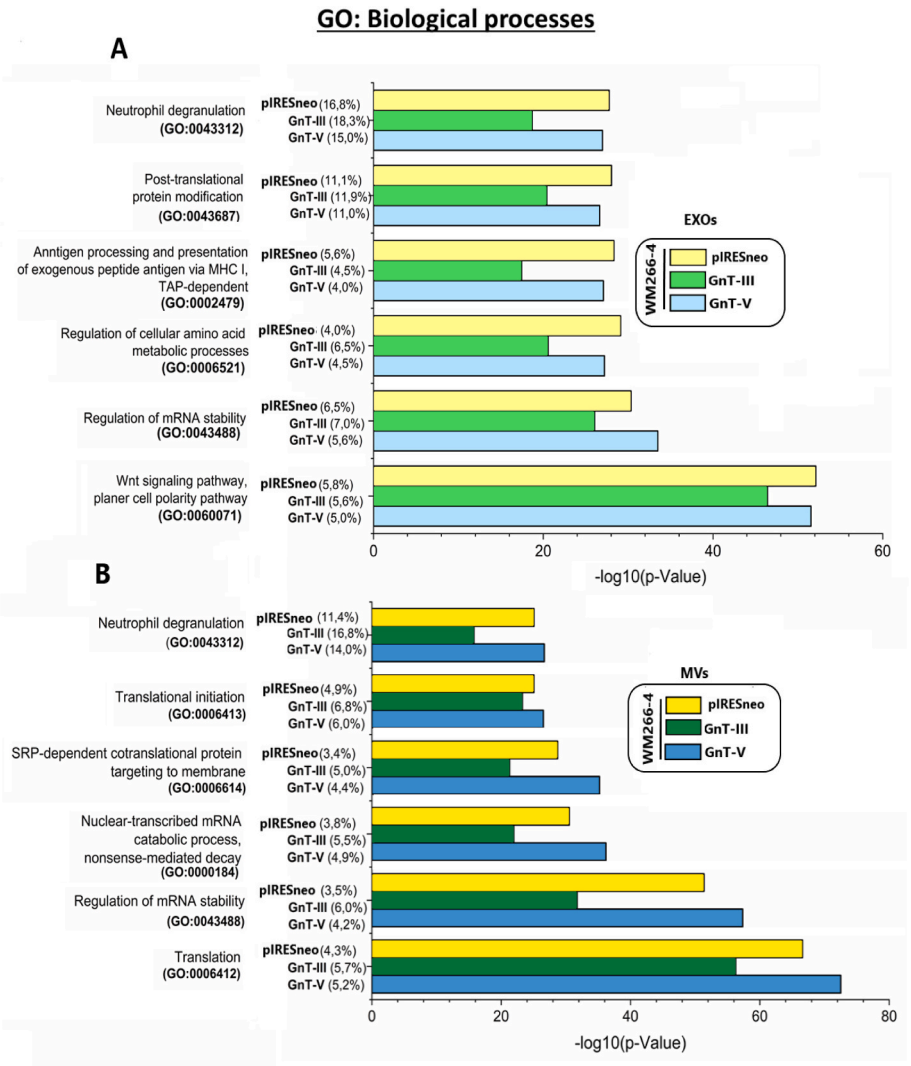



**Fig. 8.** Gene ontology (GO) analysis of proteins identified in EXOs (A) and MVs (B) derived from WM266-4$^{pIRESneo}$, WM2664$^{GnT-III}$, and WM266-4$^{GnT-V}$ performed with the use of FunRich 2.0 software with UniProt database as a reference. For the GO term "biological processes", six categories with the highest statistical significance of protein enrichment within the specific category ($p < 0.001$) are presented as graphs. The percentages in parentheses represent the percentage of proteins identified for a given GO category out of all proteins identified in a given population of EXOs or MVs. Complete results of GO analysis are provided in Supplementary Data 2.

Therefore, it can be assumed that *MGAT3* overexpression contributes to the altered sorting of these proteins to EXOs.

In addition, LFQ analysis showed that 28 and 38 proteins present in WM266-4$^{GnT-V}$-derived EXOs were up- and downregulated, respectively, compared to WM266-4$^{pIRESneo}$-derived EXOs (Table 3). In WM266-4$^{GnT-V}$-derived EXOs, one of the most abundant protein was GnT-V, which was overexpressed in WM266-4$^{GnT-V}$ cells. Overexpression of *MGAT5* in melanoma cells is associated with an increase in metastatic capacity, therefore, the presence of the protein in exosomes indicates the ability to transfer the metastatic phenotype to less malignant cells [78]. Apolipoprotein M (ApoM) was also more abundant in EXOs derived from both WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells compared to EXOs derived from WM266-4$^{pIRESneo}$ cells. There are no studies on the role of ApoM in melanoma. However, ApoM has been shown to promote proliferation and invasion in non-small cell lung cancers via ERK1/2 and PI3K/AKT signaling pathways [79]. In turn, overexpression of ApoM inhibited the proliferation, migration, and invasion of hepatoma cells [80]. Moreover, mimecan, also overexpressed in WM266-4$^{GnT-V}$-derived EXOs, reverses epithelial to mesenchymal transition and invasiveness in colorectal cancer via EGFR/Akt [81]. The tumor-suppressive role of mimecan has also been observed in melanoma. Overexpression of mimecan increased stress-triggered autophagy-induced cell death following ER stress in mouse melanoma cell lines [82].

Another upregulated protein, sorting nexin 2 (SNX2), is involved in several steps of intercellular trafficking by interacting with phosphatidylinositol 3-phosphate-containing membranes [83]. SNX2 plays a key role in the membrane trafficking of growth factor receptors, e.g. epidermal growth factor receptor (EGFR) or c-Met. Studies have shown that silencing of SNX2 alters the sensitivity to anticancer drugs targeting c-Met (SU11274) and EGFR (gefitinib and erlotinib) in lung cancer [84]. This is another example of EVs transferring a phenotype associated with drug resistance. In terms of downregulated proteins, VEGFR1, transferrin receptor protein 1, and cell migration-inducing and hyaluronan-binding protein (CEMIP) were identified as proteins with reduced abundance in EXOs derived from WM266-4$^{GnT-V}$ cells compared to WM266-4$^{pIRESneo}$-derived EXOs. BRAFv600E melanomas are sensitive to BRAF inhibitors such as vemurafenib. However, drug resistance to BRAF inhibitors often requires the use of mitogen-activated protein kinase (MEK) inhibitors. Overexpression of endosomal CEMIP has been shown to cause resistance to MEK inhibitors in colorectal cancer [85]. Furthermore, CEMIP induces immune evasion of colorectal

**Table 2**
Involvement of selected proteins present in EXOs and MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells in glycosylation.

| Gene | Protein | Function | pIRESneo MVs | pIRESneo EXO | GnT-V MVs | GnT-V EXO | GnT-III MVs | GnT-III EXO |
|---|---|---|---|---|---|---|---|---|
| *ST3GAL4* | ST3 β-galactoside α-2,3-sialyltransferase 4 | Terminal sialylation of glycoproteins and glycolipids | + | + | – | + | + | – |
| *GALNT2* | polypeptide N-acetylgalactosaminyltransferase 2 | Transfer of an N-acetyl-D-galactosamine residue to a serine or threonine residue on the protein receptor | + | + | + | + | – | + |
| *PMM2* | phosphomannomutase 2 | Synthesis of the GDP-mannose and dolichol-phosphate-mannose | + | – | – | – | – | – |
| *DDOST* | dolichyl-diphosphooligosaccharide—protein glycosyltransferase 48 kDa subunit | Catalyses the initial transfer of $Glc_3Man_9GlcNAc_2$ from the lipid carrier dolichol-pyrophosphate to an asparagine residue | + | – | – | – | – | – |
| *MAN2A1* | α-mannosidase 2 | Catalyses the first committed step in the biosynthesis of complex type N-glycans; controls conversion of oligomannose type to complex type N-glycans | + | – | – | – | – | – |
| *PGM3* | phosphoacetylglucosamine mutase | Catalyses the conversion of GlcNAc-6-P into GlcNAc-1-P during the synthesis of uridine diphosphate/UDP-GlcNAc | + | – | – | – | – | – |
| *RPN2* | dolichyl-diphosphooligosaccharide—protein glycosyltransferase subunit 2 | Subunit of the oligosaccharyl transferase (OST) complex that catalyses the initial transfer of $Glc_3Man_9GlcNAc_2$ | + | – | + | – | – | – |
| *GFPT1* | glucosamine-fructose-6-phosphate aminotransferase isomerizing 1 | Involved in regulating the availability of precursors for N- and O-linked glycosylation of proteins | + | + | + | + | + | + |
| *MGAT5* | α-1,6-mannosylglycoprotein 6-β-N-acetylglucosaminyltransferase A | Catalyses the addition of GlcNAc in β1-6 linkage to the α-linked mannose of biantennary N-glycans | + | + | + | + | + | + |
| *UGGT1* | UDP-glucose:glycoprotein glucosyltransferase | Reglucosylates single N-glycans near the misfolded part of the protein, thus providing quality control for protein folding in the endoplasmic reticulum | + | + | + | + | – | + |
| *MAN2B1* | mannosidase α class 2B member 1 | Necessary for the catabolism of N-linked carbohydrates released during glycoprotein turnover | + | – | + | + | – | – |
| *MGAT1* | α-1,3-mannosyl-glycoprotein 2-β-N-acetylglucosaminyltransferase | Essential for the conversion of oligomannose to hybrid and complex N-glycans | – | + | + | + | – | – |
| *MGAT4B* | α-1,3-mannosyl-glycoprotein 4-β-N-acetylglucosaminyltransferase B | Glycosyltransferase that catalyses the transfer of UDP-GlcNAc through a β1-4 linkage and participates in the synthesis of tri- and tetra-antennary N-glycans | – | + | – | + | – | – |
| *MANBA* | β-mannosidase A | Cleaves the single β-linked mannose residue from the non-reducing end of all N-linked glycoprotein oligosaccharides | – | – | – | + | – | – |

cancer via internalization and degradation of major histocompatibility complex I (MHC-I) [86]. Therefore, CEMIP may potentially be a cause of treatment failure with BRAF and MEK inhibitors in melanoma.

Similar to the proteins present in EXOs derived from WM266-4$^{GnT-III}$ cells, the top 10 up- and downregulated proteins present in EXOs derived from *MGAT5* overexpressing WM266-4 cells were analyzed with glycosylation-related databases (Table 5). The trends in the number of N-glycosylation sites of upregulated proteins were similar to EXOs derived from WM266-4$^{GnT-III}$ cells. As for upregulated proteins, mimecan possessed the highest number of N-glycosylation sites. However, out of the three reported N-glycosylation sites, two of them were reported as Asn-carrying N-glycans. Proteins downregulated in EXOs derived from WM266-4$^{GnT-V}$ cells possessed up to 9 reported N-glycosylation sites. Interestingly, neuroblast differentiation-associated protein AHNAK has a total of 104 reported glycosylation sites, but only one of them (Asn5718) is associated with N-glycosylation. Finally, it should be mentioned that even if the proteins present in EXOs derived from WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells possess various known reported N-glycosylation sites and sites with documented N-glycan presence, information on N-glycan types on Asn and exact sugar compositions of glycans is still not available. However, the fact that overexpression of both *MGAT3* and *MGAT5* led to a reduced abundance of proteins with more reported N-glycosylation sites and Asn with documented presence of N-glycan supports a role for complex type N-glycans in protein sorting into EXOs. It can also be hypothesized that some proteins are sorted into EXOs only when they possess β-1,6 branching or bisected GlcNAc. Changes in the exosomal cargo composition of cells overexpressing *MGAT3* or *MGAT5* may therefore result from different sorting efficiency of proteins with and without a defined sugar structure, as well as from sorting of the same protein with a different glycosylation state.

*3.3.2.2. iBAQ quantitative proteomic analyses for MVs samples.* As mentioned earlier, due to the large differences in the number of identified proteins between individual MV samples, the iBAQ method was used for qualitative analysis. The fifty most abundant proteins in the analyzed MVs were presented in Table 6. Strikingly, a lower variability of protein expression was observed in the three tested genetic variants of MVs compared to the quantitative changes seen in EXOs.

One of the most abundant proteins present in MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells was apolipoprotein E (ApoE). The germline ApoE genotype has been shown to be a predictive factor in human melanoma. The ApoE4 and ApoE2 allele resulted in prolonged and reduced survival, respectively. The ApoE4 variant has been shown to suppress melanoma progression by enhancing anti-tumor immunity [87]. In addition, ApoE plays an immune-suppressive role in various types of tumors. Serum ApoE levels gradually increase during the growth of B16–F10 melanoma cells, and ApoE-secreting melanoma cells suppress T-cell activation in vitro. However, this suppressive effect is not seen in conditioned media from

**Table 3**
Quantitative comparison of proteins present in WM266-4$^{pIRESneo}$-derived EXOs and WM266-4$^{GnT-III}$- or WM266-4$^{GnT-V}$-derived EXOs based on LFQ. The ten most highly upregulated and downregulated proteins in WM266-4$^{GnT-III}$-derived EXOs compared to WM266-4$^{pIRESneo}$-derived EXOs are presented by fold change values. Additionally, q values obtained by Student's T test are given. Full results are available in Supplementary Data 2.

| Proteins upregulated in WM266-4$^{GnT-III}$ EXOs | | | Proteins downregulated in WM266-4$^{GnT-IIII}$ EXOs | | |
|---|---|---|---|---|---|
| Protein name (Gene) | Fold change value | Student's T test q-value | Protein name (Gene) | Fold change value | Student's T test q-value |
| Lipocalin-1 *(LCN1)* | 17.86 | 0.0241 | Phospholipase A1 member A *(PLA1)* | −12.66 | 0.0238 |
| Desmocollin-1 *(DSC1)* | 15.44 | 0.0448 | Tetraspanin-4 *(TSPAN4)* | −13.41 | 0.0254 |
| Serpin B3 *(SERPINB3)* | 14.84 | 0.0203 | UDP-glucose glycoprotein glucosyltransferase 1 *(UGGT1)* | −15.24 | 0.0461 |
| Serpin B12 *(SERPINB12)* | 13.10 | 0.0319 | VPS10 domain-containing receptor SorCS1 *(SORCS1)* | −15.28 | 0.0484 |
| Desmoglein-1 *(DSG1)* | 12.69 | 0.0138 | Bone morphogenetic protein 1 *(BMP1)* | −15.48 | 0.000 |
| Arginase-1 *(ARG1)* | 11.63 | 0.0414 | Immunoglobulin superfamily member 3 *(IGSF3)* | −18.32 | 0.0149 |
| Apolipoprotein M *(APOM)* | 11.23 | 0.0498 | Integrin alpha-2 *(ITGA2)* | −19.38 | 0.0062 |
| Reticulocalbin-1 *(RCN1)* | 10.86 | 0.0135 | Alpha-2-macroglobulin *(A2M)* | −24.77 | 0.0062 |
| Small nuclear ribonucleoprotein Sm D3 *(SNRPD3)* | 10.52 | 0.0322 | Extracellular superoxide dismutase [Cu–Zn] *(SOD3)* | −60.62 | 0.0051 |
| Catalase *(CAT)* | 9.94 | 0.0131 | FRAS1-related extracellular matrix protein 1 *(FREM1)* | −79.28 | 0.0490 |

| Proteins upregulated in WM266-4$^{GnT-V}$ EXOs | | | **Proteins downregulated in WM266-4$^{GnT-V}$EXOs** | | |
|---|---|---|---|---|---|
| Protein name | **Fold change value** | **Student's T test q-value** | **Protein name** | **Fold change value** | **Student's T test q-value** |
| Apoplipoprotein M *(APOM)* | 49.91 | 0.000 | Proprotein convertase subtilisin/kexin type 6 *(PCSK6)* | −6.20 | 0.0000 |
| α-1,6-mannosylglycoprotein 6-β-N-acetylglucosaminyltransferase A *(MGAT5)* | 11.83 | 0.0032 | DBH-like monooxygenase protein 1 *(MOXD1)* | −6.29 | 0.0197 |
| Small nuclear ribonucleoprotein Sm D2 *(SNRPD2)* | 20.23 | 0.0166 | Neuroblast differentiation-associated protein AHNAK *(AHNAK)* | −6.36 | 0.0040 |
| Mimecan *(OGN)* | 6.38 | 0.0194 | Peptidyl-glycine alpha-amidating monooxygenase *(PAM)* | −6.67 | 0.0047 |
| Alcohol dehydrogenase [NADP(+)] *(ADH)* | 5.81 | 0.0172 | Vascular endothelial growth factor receptor 1 *(VEGFR1)* | −6.74 | 0.0191 |
| Sorting nexin-2 *(SNX2)* | 5.70 | 0.0095 | Out at first protein homolog *(OAF)* | −7.00 | 0.0222 |
| General vesicular transport factor p115 *(USO1)* | 5.38 | 0.0051 | Tumor necrosis factor-inducible gene 6 protein *(TSG6)* | −7.10 | 0.0027 |
| 26S protease regulatory subunit 7 *(PSMC2)* | 4.68 | 0.0057 | Collagen alpha-1(XII) chain *(COL12A1)* | −7.86 | 0.0029 |
| Ubiquitin-conjugating enzyme E2 variant 1 *(UBE2V1)* | 4.65 | 0.0058 | Transferrin receptor protein 1 *(TFRC)* | −8.90 | 0.0033 |
| FACT complex subunit SPT16 *(SUPT16H)* | 3.35 | 0.0356 | Cell migration-inducing and hyaluronan-binding protein *(CEMIP)* | −9.15 | 0.0041 |

ApoE knock-out tumor cells [88].

Moreover, increased levels of proteinase inhibitors, *e.g.* glia-derived nexin (GDN), are associated with the development melanoma metastases. GDN expression has been associated with sentinel lymph node metastasis in primary cutaneous melanoma [89]. Furthermore, GDN has been found to contribute to the highly invasive potential of slow-cycling melanoma cells, also known as label-retaining cells [90]. In the present study, GDN was also one of the highest expressed proteins in all MV populations. In addition, L-lactate dehydrogenase (LDH) has been identified as one of the most highly expressed MV proteins in the presented study. Overexpression of LDH is a well-known marker of malignancy in melanoma [91]. Large amounts of lactate are synthesized by cancer cells in response to hypoxia or pseudohypoxia, which impairs the anti-tumor immune response, also in melanoma [92]. It has been proposed that targeting LDH and other molecules involved in lactate metabolism may improve the efficacy of immuno-therapies. Therefore, targeting vesicular LDH transport may positively impact melanoma treatment. Sorting of LDH into MVs, independent of *MGAT3* and *MGAT5* expression in melanoma cells, suggests that vesicular LDH is a molecule with high diagnostic/prognostic potential.

Increased vesicular transport of other enzymes of basal metabolism, *i.e.* α-enolase (ENO1), was also characteristic of all three MV populations. ENO1 has been found to be overexpressed in melanoma tissues and promotes invasion, migration and proliferation of tumor cells [93]. Like LDH, ENO1 also increases lactate levels in melanoma tissues. In contrast to pro-tumor proteins present in MVs, annexin A5 (ANXA5) has been identified to possess anti-tumor properties in melanoma. ANXA5 treatment significantly reduced the size of murine melanoma xenograft, while its reduced expression correlated with increased VEGF expression [94]. However, the presence of ANXA5 in MVs can reduce the effect of therapeutic antibodies directed against phosphatidylserine (PS). Studies of PS-targeting mAbs with immune checkpoint blockade (anti-CTLA-4 or anti-programmed death receptor 1 (PD-1)) have been evaluated in breast cancer and melanoma, indicating future potential for improving immune checkpoint therapy [95]. However, the presence of ANXA5 in MV cargo may interfere with this therapy via annexin-binding PS.

Proteins present in MVs were also analyzed using glycosylation-related databases (Table 7). Compared to the analogous data for EXOs (Table 5), the data for proteins identified in MVs was more complex. Most of the proteins analyzed in this part of research showed different abundancy in MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells (Table 6). Apolipoprotein E and actin 2 were highly abundant in all MVs, these two proteins do not possess any N-glycosylation sites and it can be assumed that N-glycans are not involved in their sorting into MVs. Fibronectin, which abundancy was higher in MVs derived from WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells compared to WM266-4$^{pIRESneo}$-derived MVs, possesses 25 glycosylation sites, including N-glycosylation sites. Asn528, Asn542, and Asn1007 of fibronectin carry complex type of glycans. Therefore, it can be assumed that β-1,6-branching of fibronectin glycans may have a positive effect on its sorting to MVs, compared to N-glycans with bisecting GlcNAc. Furthermore, tenascin, which was found to be one of the most abundant

**Table 4**
Information on the glycosylation status of the top 10 down- or upregulated proteins WM266-4$^{GnT-III}$-derived EXOs compared to WM266-4$^{pIRESneo}$-derived EXOs based on the GlyGen database and GlyConnect database.

| Protein (Gene) | Up- or down regulated in EXOs | Reported glycosylation sites | Reported sites with N-glycans | Type of N-glycans on known Asn (GlyConnect) |
|---|---|---|---|---|
| Lipocalin 1 *(LCN1)* | UP | 1 | 0 | N/A |
| Desmocollin-1 *(DSC1)* | UP | 8 | 0 | Undefined |
| Serpin B3 *(SERPIN3B)* | UP | 2 | 0 | N/A |
| Serpin B12 *(SERPINB12)* | UP | 0 | 0 | N/A |
| Desmoglein-1 *(DSG1)* | UP | 3 | Asn10 | Oligomannose |
| Arginase-1 *(ARG1)* | UP | 1 | N/A | N/A |
| Apolipoprotein M *(APOM)* | UP | 1 | Asn135 | Undefined |
| Reticulocalbin-1 *(RCN1)* | UP | 5 | Asn53 | Undefined |
| Small nuclear ribonucleoprotein Sm D3 *(SNRPD3)* | UP | 1 | 0 | N/A |
| Catalase *(CAT)* | UP | 6 | Asn244, Asn439, Asn481 | Undefined |
| Phospholipase A1 member A *(PLA1)* | DOWN | 1 | 0 | N/A |
| Tatraspanin-4 *(TSPN4)* | DOWN | 0 | 0 | N/A |
| UDP-glucose glycoprotein glucosyltransferase 1 *(UGGT1)* | DOWN | 7 | Asn269, Asn536, Asn1228 | Undefined |
| VPS10 domain-containing receptor SorCS1 *(SORC1)* | DOWN | 5 | Asn256, Asn286, Asn920 | Undefined |
| Bone morphogenetic protein 1 *(BMP1)* | DOWN | 8 | Asn91, Asn142, Asn363, Asn599 | Undefined |
| Immunoglobulin superfamily member 3 *(IGSF3)* | DOWN | 9 | Asn43, Asn101, Asn655, Asn700, Asn825, Asn842, Asn961, Asn1077 | Undefined |
| Integrin α2 *(ITGA2)* | DOWN | 10 | Asn105, Asn112, Asn343, Asn432, Asn460, Asn475, Asn699, Asn1057, Asn1074 | Undefined |
| α-2-macroglobulin *(A2M)* | DOWN | 14 | Asn55, Asn247, Asn396, Asn410, Asn869, Asn991, Asn1424 | Asn869, Asn1424 - complex type |
| Extracellular superoxide dismutase [Cu–Zn] *(SOD3)* | DOWN | 2 | Asn107 | Undefined |
| FRAS1-related extracellular matrix protein 1 *(FREM1)* | DOWN | 2 | Asn1683 | Undefined |

**Table 5**
Information on the glycosylation status of the top 10 down- or upregulated proteins in WM266-4$^{GnT-V}$-derived EXOs compared to WM266-4$^{pIRESneo}$-derived EXOs based on the GlyGen database and GlyConnect database.

| Protein (Gene) | Up- or down regulated in EXOs | Reported glycosylation sites | Reported sites with N-glycans | Type of N-glycans on known Asn (GlyConnect) |
|---|---|---|---|---|
| Apolipoprotein M *(APOM)* | UP | 1 | Asn135 | Undefined |
| α-1,6-mannosylglycoprotein 6-β-N-acetylglucosaminyltransferase A *(MGAT5)* | UP | 2 | Asn107 | Undefined |
| Small nuclear ribonucleoprotein Sm D2 *(SNRPD2)* | UP | 3 | Asn38 | Undefined |
| Mimecan *(OGN)* | UP | 4 | Asn214, Asn258 | Undefined |
| Alcohol dehydrogenase [NADP(+)] *(ADH)* | UP | 1 | 0 | N/A |
| Sorting nexin-2 *(SNX2)* | UP | 2 | 0 | N/A |
| General vesicular transport factor p115 *(USO1)* | UP | 1 | 0 | N/A |
| 26S protease regulatory subunit 7 *(PSMC2)* | UP | 2 | Asn379 | Undefined |
| Ubiquitin-conjugating enzyme E2 variant 1 and 2 *(UBE2V1)* | UP | 1 | 0 | N/A |
| FACT complex subunit SPT16 *(SUPT16H)* | UP | 4 | 0 | N/A |
| Proprotein convertase subtilisin/kexin type 6 *(PCSK6)* | DOWN | 3 | Asn914, Asn932 | Undefined |
| DBH-like monooxygenase protein 1 *(MOXD1)* | DOWN | 2 | Asn114, Asn517 | Undefined |
| Neuroblast differentiation-associated protein AHNAK *(AHNAK)* | DOWN | 104 (103 sites of O-linked glycans) | Asn5718 | Undefined |
| Peptidyl-glycine alpha-amidating monooxygenase *(PAM)* | DOWN | 8 | 0 | N/A |
| Vascular endothelial growth factor receptor 1 *(VEGFR1)* | DOWN | 10 | Asn100, Asn164, Asn196, Asn251, Asn323, Asn402, Asn474, Asn547, Asn625 | Undefined |
| Out at first protein homolog *(OAF)* | DOWN | 3 | 0 | N/A |
| Tumor necrosis factor-inducible gene 6 protein *(TNFAIP6)* | DOWN | 2 | Asn118, Asn258 | N/A |
| Collagen alpha-1(XII) chain *(COL12A1)* | DOWN | 19 | Asn1763, Asn2206, Asn2528, Asn2679 | Undefined |
| Transferrin receptor protein 1 *(TFRC)* | DOWN | 12 | Asn148, Asn251, Asn317, Asn372, Asn615, Asn727 | Asn727 – oligomannose type |
| Cell migration-inducing and hyaluronan-binding protein *(CEMIP)* | DOWN | 3 | Asn165, Asn889 | Undefined |

protein in MVs derived from WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells, is the protein with 15 N-glycosylation sites, nine of which are occupied by complex type N-glycans. In the case of tenascin, it can be assumed that β-1,6-branching of complex type N-glycans may play an important role in the sorting of tenascin to MVs. Finally, galectin-3-binding protein (G3BP), which is highly abundant in MVs derived from both WM266-

**Table 6**
Fifty most abundant proteins (Top50) detected in MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells. Intensity Rank was based on mean iBAQ values. Relevant gene names are provided in brackets.

| Intensity Rank | WM266-4$^{pIRESneo}$-derived MVs | WM266-4$^{GnT-III}$-derived MVs | WM266-4$^{GnT-V}$-derived MVs |
|---|---|---|---|
| 1 | Actin, cytoplasmic 2 *(ACTB)* | Apolipoprotein E *(APOE)* | Apolipoprotein E *(APOE)* |
| 2 | Apolipoprotein E *(APOE)* | Glia-derived nexin *(GLDN)* | Actin, cytoplasmic 2 *(ACTB)* |
| 3 | Annexin A2 *(ANXA2)* | Actin, cytoplasmic 2 *(ACTB)* | Glia-derived nexin *(GLDN)* |
| 4 | Glyceraldehyde-3-phosphate dehydrogenase *(GAPDH)* | Annexin A2 *(ANXA2)* | Fibronectin *(FN1)* |
| 5 | Peptidyl-prolyl *cis*-trans isomerase A *(PPIA)* | Fibronectin *(FN1)* | Glyceraldehyde-3-phosphate dehydrogenase *(GAPDH)* |
| 6 | Glia-derived nexin *(GLDN)* | Annexin A5 *(ANXA5)* | Annexin A2; Putative annexin A2-like protein *(ANXA2)* |
| 7 | Lactadherin *(MFGE8)* | Glyceraldehyde-3-phosphate dehydrogenase *(GAPDH)* | Alpha-enolase *(ENO1)* |
| 8 | Alpha-enolase *(ENO1)* | Alpha-enolase *(ENO1)* | Annexin A5 *(ANXA5)* |
| 9 | Annexin A5 *(ANXA5)* | Growth/differentiation factor 15 *(GDF15)* | Peptidyl-prolyl *cis*-trans isomerase A *(PPIA)* |
| 10 | Cofilin-1 *(CFL1)* | Sushi repeat-containing protein SRPX *(SRPX)* | Lactadherin *(MFGE8)* |
| 11 | L-lactate dehydrogenase A chain *(LDHA)* | Peptidyl-prolyl *cis*-trans isomerase A *(PPIA)* | L-lactate dehydrogenase A chain *LDHA* |
| 12 | Moesin *(MSN)* | Pyruvate kinase PKM *(PKM)* | Pyruvate kinase PKM *(PKM)* |
| 13 | Pyruvate kinase PKM *(PKM)* | L-lactate dehydrogenase A chain *(LDHA)* | Tenascin *(TNC)* |
| 14 | 14-3-3 protein epsilon *(YWHAE)* | Syntenin-1 *(SDCBP)* | Cofilin-1 *(CFL1)* |
| 15 | Profilin-1 *(PFN1)* | Tenascin *(TNC)* | 14-3-3 protein epsilon *(YWHAE)* |
| 16 | Ubiquitin-40S ribosomal protein S27a *(RPS27A)* | Cofilin-1 *(CFL1)* | Galectin-3-binding protein *(LGALS3BP)* |
| 17 | Elongation factor 1-alpha 1 *(EEF1A1)* | Galectin-3-binding protein *(LGALS3BP)* | Ubiquitin-40S ribosomal protein S27a *(RPS27A)* |
| 18 | 14-3-3 protein zeta/delta *(YWHAZ)* | 14-3-3 protein epsilon *(YWHAE)* | Syntenin-1 *(SDCBP)* |
| 19 | Fructose-bisphosphate aldolase A *(ALDOA)* | Catenin beta-1 *(CTNNB1)* | Heat shock protein HSP 90-beta *(HSP90B1)* |
| 20 | Sodium/potassium-transporting ATPase subunit alpha-1 *(ATP1A1)* | Heat shock protein HSP 90-beta *(HSP90B1)* | Fructose-bisphosphate aldolase A *(ALDOA)* |
| 21 | Heat shock protein HSP 90-beta *(HSP90B1)* | Ubiquitin-40S ribosomal protein S27a *(RPS27A)* | Growth/differentiation factor 15 *(GDF15)* |
| 22 | Guanine nucleotide-binding protein G(I)/G(S)/G(O) subunit gamma-12 *(GNG12)* | Elongation factor 1-alpha 1 *(EEF1A1)* | Moesin *(MSN)* |
| 23 | Heat shock cognate 71 kDa protein *(HSPA8)* | Annexin A1 *(ANXA1)* | Elongation factor 1-alpha 1 *(EEF1A1)* |
| 24 | Syntenin-1 *(SDCBP)* | Heat shock cognate 71 kDa protein *(HSPA8)* | Sodium/potassium-transporting ATPase subunit alpha-1 *(ATP1A1)* |
| 25 | Fibronectin *(FN1)* | Dermcidin *(DERM)* | Tubulin alpha-1A chain *(TUBA1A)* |
| 26 | Guanine nucleotide-binding protein G(i) subunit alpha-2 *(GNAI2)* | 14-3-3 protein zeta/delta *(YHWAZ)* | Tubulin beta-4B chain *(TUBB4B)* |
| 27 | Tubulin alpha-1A chain; Tubulin alpha-3E chain *(TUBA1A/TUBA3E)* | Tubulin alpha-1A chain *(TUBB4B)* | Heat shock cognate 71 kDa protein *(HSPA8)* |
| 28 | Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1 *(GNB1)* | Fructose-bisphosphate aldolase A *(ALDOA)* | Profilin-1 *(PFN1)* |
| 29 | Tubulin beta-4B chain *(TUBB4B)* | Tubulin beta-4B chain *(TUBB4B)* | CD9 antigen *(CD9)* |
| 30 | Chloride intracellular channel protein 1 *(CLIC1)* | Phosphoglycerate kinase 1 *(PGK1)* | Guanine nucleotide-binding protein G (I)/G(S)/G(O) subunit gamma-12 *(GNG12)* |
| 31 | CD9 antigen *(CD9)* | Guanine nucleotide-binding protein G(I)/G(S)/G(O) subunit gamma-12 *(GNG12)* | Guanine nucleotide-binding protein G (I)/G(S)/G(T) subunit beta-1 *(GNB1)* |
| 32 | Transforming protein RhoA *(RHOA)* | CD9 antigen *(CD9)* | Phosphoglycerate kinase 1 *(PGK1)* |
| 33 | Vimentin *(VIM)* | Profilin-1 *(PFN1)* | Guanine nucleotide-binding protein G(i) subunit alpha-2 *(GNAI2)* |
| 34 | Ras-related C3 botulinum toxin substrate 1 *(RAC1)* | Synaptic vesicle membrane protein VAT-1 homolog *(VAT1)* | Galectin-3 *(LGALS3)* |
| 35 | Galectin-3 *(LGALS3)* | Apolipoprotein D *(APOD)* | L-lactate dehydrogenase B chain *(LDHB)* |
| 36 | Phosphoglycerate kinase 1 *(RGK1)* | SPARC *(SPARC)* | 14-3-3 protein zeta/delta *(YHWAZ)* |
| 37 | ADP-ribosylation factor 1 *(ARF1)* | 72 kDa type IV collagenase; PEX *(MMP2)* | Gelsolin *GSN)* |
| 38 | L-lactate dehydrogenase B chain *(LDH8)* | L-lactate dehydrogenase B chain *(LDHB)* | Triosephosphate isomerase *(TPI1)* |
| 39 | Chloride intracellular channel protein 4 *(CLIC4)* | Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-1 *(GNB1)* | Vimentin *(VIM)* |
| 40 | Chondroitin sulfate proteoglycan 4 *(CSPG4)* | Guanine nucleotide-binding protein G(i) subunit alpha-2 *(GNAI2)* | Sushi repeat-containing protein SRPX *(SRPX)* |
| 41 | Sodium/potassium-transporting ATPase subunit beta-3 *(ATP1B3)* | Vimentin *(VIM)* | 72 kDa type IV collagenase; PEX *(MMP2)* |
| 42 | Ras-related protein Rap-1b *(RAP1B)* | Galectin-3 *(LGALS3)* | Synaptic vesicle membrane protein VAT-1 homolog *(VAT1)* |
| 43 | Gelsolin *(GSN)* | Sodium/potassium-transporting ATPase subunit alpha-1 *(ATP1A1)* | Lysyl oxidase homolog 4 *(LOXL4)* |
| 44 | Sushi repeat-containing protein SRPX *(SRPX)* | Insulin-like growth factor-binding protein 3 *(IGFBP3)* | Annexin A1 *(ANXA1)* |

(*continued on next page*)

**Table 6** (*continued*)

| Intensity Rank | WM266-4$^{pIRESneo}$-derived MVs | WM266-4$^{GnT-III}$-derived MVs | WM266-4$^{GnT-V}$-derived MVs |
|---|---|---|---|
| 45 | Guanine nucleotide-binding protein G(I)/G(S)/G(T) subunit beta-2 *(GNB2)* | Gelsolin *(GSN)* | Ras-related C3 botulinum toxin substrate 1 *(RAC1)* |
| 46 | 4F2 cell-surface antigen heavy chain *(SLC3A2)* | Lysyl oxidase homolog 4 *(LOXL4)* | Insulin-like growth factor-binding protein 3 *(IGFBP3)* |
| 47 | Synaptic vesicle membrane protein VAT-1 homolog *(VAT1)* | Triosephosphate isomerase *(TPI1)* | Prostaglandin F2 receptor negative regulator *(PTGFRN)* |
| 48 | Myosin regulatory light chain 12B *(MYL12B)* | Dual specificity mitogen-activated protein kinase kinase 1 *(MAP2K1)* | 4F2 cell-surface antigen heavy chain *(SLC3A2)* |
| 49 | 14-3-3 protein beta/alpha *(YHWAB)* | C-type lectin domain family 11 member A *(CLEC11A)* | Transforming protein RhoA *(RHOA)* |
| 50 | Ras-related protein Rab-7a *(RAB7A)* | Nucleophosmin *(NPM1)* | Apolipoprotein D *(APOD)* |

4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells has three sites with complex type structures, one with hybrid type structure, and one with high mannose structure. Therefore, β-1,6-branching and presence of bisecting GlcNAc may also play a role in sorting G3BP into MVs, as G3BP did not show such a high abundancy in WM266-4$^{pIRESneo}$-derived MVs.

In summary, the presented quantitative proteomic data indicates that changes in the expression of *MGAT3* and *MGAT5* in the genetic variants of WM266-4 human melanoma cells affect the protein content of EVs released by these cells. Although within top 10 most abundant proteins there was little diversity regarding their iBAQ-based intensity ranking, less abundant proteins showed more pronounce differences. For instance, fibronectin was ranked 25th, 4th and 5th, profilin as 15th, 33rd and 28th, sushi repeat-containing protein SRPX as 44th, 10th and 40th, or sodium/potassium-transporting ATPase subunit alpha-1 as 20th, 43rd and 24th in MVs from WM266-4$^{pIRESneo}$and WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, respectively. More explicit changes to the protein content were observed for EXOs compared to MVs. This may be related to differences in the biogenesis of these two distinct EV populations. MVs are formed within the outer cell membrane and thus contain mainly membrane/transmembrane proteins. On the other hand, EXOs are formed within multivesicular bodies (from late endosomal compartments). Therefore, EXOs may contain more intracellular proteins, including those sorted directly from the endoplasmic reticulum and the Golgi apparatus, which are sites of glycan synthesis.

### 3.4. Diagnostic potential of melanoma-derived EVs

To analyze the diagnostic and prognostic potential of proteins identified in EXOs and MVs derived from three genetic variants of WM266-4 cells, various databases, *i.e.*, The Human Protein Atlas, DysGenNET, KEGG, DISEASES, STRING, GlyConnect, and GlyGen. GlyConnect and GlyGen databases, were used to analyze the glycosylation of proteins with statistically significant prognostic value in cutaneous melanoma based on The Human Protein Atlas data. KEGG and DISEASES were used to check whether proteins typical for melanomagenesis were present in EVs.

The analysis of the predictive potential of proteins present in EXOs and MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells was performed using the data included in The Human Protein Atlas. The list of proteins common to all EXO samples and all MV samples was searched through the list of proteins with favorable or unfavorable prognostic value in melanoma patients. Based on the data provided by The Human Protein Atlas, 162 proteins were classified as

**Table 7**
Basic information on N-glycosylation profile of the top 20 proteins present in MVs derived from WM266-4$^{pIRESneo}$, WN266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells with the highest mean log2(iBAQ) values. The tables were prepared using glyconnect.expasy.org and glygen.org databases.

| Protein | Number of reported glycosylation sites | Reported Asn with N-glycans | Asn with known oligomannose/hybrid/complex structures (GlyConnect) |
|---|---|---|---|
| Actin, cytoplasmic 2 *(ACTB)* | 10 | Asn12 | Undefined |
| Apolipoprotein E *(APOE)* | 8 | 0 | N/A |
| Annexin A2 *(ANXA2)* | 2 | 0 | N/A |
| Glyceraldehyde-3-phosphate dehydrogenase *(GAPDH)* | 4 | Asn149, Asn239 | Undefined |
| Peptidyl-prolyl *cis*-*trans* isomerase A *(PPIA)* | 3 | Asn71, Asn108 | Undefined |
| Glia-derived nexin *(SERPINB1)* | 4 | Asn118, Asn159 | Undefined |
| Lactadherin *(MFGE8)* | 4 | Asn238, Asn325, Asn329 | Asn238 – oligomannose, hybrid<br>Asn325 – complex<br>Asn329 – hybrid |
| Alpha-enolase *(ENO1)* | 4 | Asn70, Asn102 | Undefined |
| Annexin A5 *(ANXA5)* | 2 | 0 | N/A |
| Cofilin-1 *(CFL1)* | 1 | 0 | N/A |
| L-lactate dehydrogenase A chain *(LDHA)* | 3 | Asn84 | N/A |
| Moesin *(MSN)* | 2 | Asn247 | N/A |
| Fibronectin *(FN1)* | 25 | Asn139, Asn430, Asn528, Asn542, Asn1007, Asn1236, Asn1291 | Asn528 – complex<br>Asn542 – complex<br>Asn1007 – complex |
| Growth/differentiation factor 15 *(GDF15)* | 6 | Asn70 | Undefined |
| Tenascin *(TNC)* | 23 | Asn38, Asn166, Asn184, Asn327, Asn1018, Asn1034, Asn1079, Asn1093, Asn1119, Asn1210, Asn1366, Asn1392, Asn1485, Asn1809 | Asn1018 – oligomannose<br>Asn38, Asn166, Asn184, Asn327, Asn1093, Asn1119, Asn1366, Asn1392, Asn1485 – complex |
| Pyruvate kinase PKM *(PKM)* | 5 | 0 | N/A |
| Galectin-3-binding protein *(LGALS3BP)* | 11 | Asn69, Asn125, Asn192, Asn398, Asn551, Asn580 | Asn69, Asn398, Asn551 – complex<br>Asn551 – hybrid<br>Asn551 – oligomannose |
| Syntenin-1 *(SDCBP)* | 1 | 0 | N/A |
| 14-3-3 protein epsilon *(YWHAE)* | 2 | Asn227 | N/A |
| HSP 90-beta *(HSP90B1)* | 4 | Asn46, Asn283, Asn389 | Undefined |

unfavorable prognostic factors in melanoma, while 45 proteins were classified as favorable factors. Table 8 shows proteins with a statistically significant p-value for the correlation of mRNA level and patient survival.

Among the proteins present in all MVs, seven proteins showed predictive potential, six of them were assigned to an unfavorable prognosis. Among the exosomal proteins, four proteins were identified in the common protein list, three of them were assigned to an unfavorable prognosis. In addition, insulin-like growth factor-binding protein 2, protein disulfide isomerase family A member 6, biglycan, and galectin 1 were identified in both EXOs and MVs. The presence of these proteins in all melanoma-derived EVs, taking into account the changes in *MGAT3* and *MGAT5* expression, gives the proteins a high potential prognostic value. However, in the future, it would be worth determining the concentration of these proteins in EVs isolated from the blood of melanoma patients to determine strict ranges of concentrations observed, among others, at different stages of melanoma development.

In the next step, the same data from The Human Protein Atlas was used to analyze individual proteins present in EXOs and MVs derived from three genetic variants of WM266-4 cells to obtain a list of prognostic factors present in distinct EV populations. Only one additional protein, hyaluronan-proteoglycan linking protein 3 (HAPLN3), was found to be a favorable factor. HAPLN3 was identified in both MVs and EXOs derived from WM266-4$^{pIRESneo}$ and WM266-4$^{GnT-V}$ cells. For proteins with an unfavorable value, 12 additional proteins were identified and are presented in Table 9. Moreover, twelve of these proteins were present in MVs, and six in EXOs. Therefore, it can be assumed that changes in the expression of the two major glycosyltransferases in melanoma development may confound the presence and abundance of proteins with prognostic potential in melanoma-derived EVs.

In further studies, the GlyConnect and GlyGen databases were searched to analyze the glycosylation profiles of all previously identified proteins with a prognostic value. Only 7 of the 20 proteins were N-glycosylated. The results of the analyses are presented in Table 10. However, these proteins possessed one to three N-glycosylation sites, and information on the N-glycan structure was very limited. To conclude, N-glycan structures of vesicular proteins with prognostic value in melanoma are rather undefined. At this point, it is difficult to assess whether the characteristic melanoma sugar structures are present on these vesicular proteins. But also, whether the glycosylation of these changes during melanoma progression and the glycosylation pattern differ from those in normal melanocytes.

Protein-Protein Interaction Networks Functional Enrichment Analysis STRING v. 12.0 (https://string-db.org) was used to identify proteins present in WM266-4-derived EVs involved in melanoma-related molecular pathways. Proteins identified in EVs were compared with the protein list in both pathways, *i.e.* the KEGG "Melanoma" molecular pathway (hsa05218) and Disease Ontology DISEASES "Melanoma" (DOID:1909) pathway. In general, MVs were more abundant in the proteins present in the KEGG and DISEASES pathway list (Table 11). The KEGG melanoma pathway was represented by 72 proteins and has protein-protein interaction (PPI) enrichment with a p-value <1,0e-16 (Fig. 9). The common proteins for the KEGG and DISEASES pathways included cyclin-dependent kinase 4 (CDK4), p53 (TP53), cyclin-dependent kinase inhibitor 1 (CDKN1A), cyclin-dependent kinase inhibitor 2A (CDKN2A), melanocyte inducing transcription factor (MITF), B-Raf (BRAF), neuroblastoma RAS viral oncogene homolog (NRAS), epidermal growth factor (EGF), and E-cadherin (CDH1). The DISEASES melanoma pathway was represented by 46 proteins and also had PPI enrichment with a p-value <1,0e-16 (Fig. 10). This possibility of identifying proteins involved in the occurrence and progression of melanoma in melanoma-derived EVs has a great diagnostic value, especially if we consider the presence of up to 10 proteins associated with these phenomena in the cargo of EVs derived from WM266-4 cells.

Finally, a search of the DisGenNET database revealed that variable numbers of proteins identified in EVs were present in the database and had GDA, DSI, DPI calculated and available number of gene-involved diseases (Table 12). The database consists of approximately 200 melanoma-related genes, whereas 24, 30, and 25 protein products encoded by these genes were present in EXOs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells, respectively. WM266-4$^{pIRESneo}$-derived MVs had 56 proteins included in the database, whereas WM266-4$^{GnT-V}$-derived MVs and WM266-4$^{GnT-III}$-derived MVs had 40 and 30 such proteins, respectively.

Some of the aforementioned proteins are closely related to melanoma, *e.g.* protein phosphatase 1 regulatory inhibitor subunit 14B (PPP1R14B) or melanotransferrin (MELTF). Other proteins mentioned, such as ubiquitin protein ligase E3 component N-Recognin 4 (UBR4), are also involved in other pathologies, including breast cancer, childhood developmental disorders, neurodevelopmental disorders, long QT syndrome, or even menopause. For example, N-myc downstream regulated 1 (NRDG1) has been associated with 29 diseases, the vast majority of which are cancer-related pathologies, *e.g.* prostate, breast, kidney, oropharyngeal and liver cancers, melanoma, non-small cell lung cancer, glioblastoma, renal cell carcinoma, head and neck carcinoma, astrocytoma, and smooth muscle tumor. However, NRDG1 was also associated

**Table 8**
Proteins present in all EXOs or MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$, and WM266-4$^{GnT-III}$ cells with favorable and unfavorable prognostic value in the diagnosis of cutaneous melanoma. The presented values of RNA expression in skin cancer cell lines in transcript per milion [nTPM], RNA expression in tissue TCGA in fragments per kilobase per million mapped fragments [FPKM], and overall survival data of melanoma patients are based on The Human Protein Atlas.

| Protein name (Gene) | RNA expression in skin cancer cell lines [nTPM] n = 62 | Cut-off level [FPKM] | Median death under cut-off [years] | Median death over cut off [year] | 3 year survival with high expression | 3 year survival with low expression | P-score of mRNA level and patient survival correlation | Presence in EVs |
|---|---|---|---|---|---|---|---|---|
| Guanine nucleotide-binding protein G(i) subunit α-1 *(GNAI1)* | 9.1 | 4.6 | 1.28 | 1.13 | 0 % | 49 % | 4.2e-5 | MVs |
| Glutathione S-transferase Mu 3 *(GSTM3)* | 5.4 | 41.51 | 1.67 | 1.01 | 0 | 45 % | 5.2e-7 | MVs |
| Insulin-like growth factor-binding protein 2 *(IGFBP2)* | 124.2 | 75.12 | 1.28 | 1.09 | 0 % | 41 % | 1.1e-4 | MVs/EXOs |
| Protein disulfide isomerase family A member 6 *(PDIA6)* | 487.5 | 163.34 | 1.52 | 1.09 | 0 % | 59 % | 2.6e-4 | MVs/EXOs |
| Selenium binding protein 1 *(SELENBP1)* | 11.4 | 34.56 | 1.98 | 1.01 | 37 % | 41 % | 3.8e-4 | MVs |
| Biglycan *(BGN)* | 4.3 | 186.75 | 1.28 | 1.01 | 0 % | 51 % | 1.4e-5 | MVs/EXOs |
| Galectin 1 *(LGALS1)* | 4788.4 | 1623.36 | 1.09 | 1.29 | 62 % | 15 % | 6.5e-4 | MVs/EXOs |

**Table 9**
Proteins present in individual EXO or MV populations derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells with favorable and unfavorable prognostic value in melanoma. Only proteins that were not abundant in all EXO and MV populations were presented. The presented values of RNA expression in skin cancer cell lines in transcript per million [nTPM], RNA expression in tissue TCGA in fragments per kilobase per million mapped fragments [FPKM], and overall survival data of melanoma patients are based on The Human Protein Atlas.

| Protein name (Gene) | RNA expression in skin cancer cell lines [nTPM] n = 62 | Cut-off level [FPKM] | Median death under cut-off [years] | Median death over cut off [years] | 3 year survival with high expression | 3 year survival with low expression | P-score of mRNA level and patient survival correlation | Presence in EVs |
|---|---|---|---|---|---|---|---|---|
| Tetraspanin 9 *(TSPAN9)* | 192 | 19.35 | 2.01 | 1.09 | 34 % | 39 % | 8.6e-4 | MVs$^{pIRESneo}$, MVs$^{GnT-V}$, EXOs$^{pIRESneo}$ |
| Regulator of chromosome condensation 2 *(RCC2)* | 59.9 | 54.56 | 1.29 | 0.99 | 0 % | 43 % | 6.8e-4 | MVs$^{pIRESneo}$, MVs$^{GnT-V}$ |
| Methionine adenosyltransferase 2A *(MAT2A)* | 188.9 | 90.17 | 1.23 | 1.17 | 0 % | 60 % | 2.4e-5 | MVs$^{pIRESneo}$ |
| Cell division cycle associated 8 *(CDCA8)* | 51.0 | 13.04 | 1.50 | 1.01 | 0 % | 48 % | 2.7e-3 | MVs$^{pIRESneo}$ |
| Malic enzyme 1 *(ME1)* | 17.8 | 3.1 | 1.29 | 1.06 | 20 % | 55 % | 8.6e-4 | MVs$^{pIRESneo}$, MVs$^{GnT-V}$, EXOs$^{GnT-V}$, EXOs$^{pIRESneo}$ |
| Complement C1q binding protein *(C1QBP)* | 376.6 | 167.2 | 1.36 | 1.01 | 0 % | 42 % | 6.7e-4 | MVs$^{pIRESneo}$, MVs$^{GnT-V}$ |
| Cullin 3 *(CUL3)* | 54.7 | 38.4 | 1.28 | 1.10 | 0 % | 52 % | 8.0e-4 | MVs$^{pIRESneo}$ |
| Leucine zipper and CTNNBIP1 domain containing *(LZIC)* | 42.3 | 22.58 | 1.29 | 1.09 | 0 % | 54 % | 0.00025 | MVs$^{pIRESneo}$ |
| C1q and TNF related 5 *(C1QTNF5)* | 1.9 | 21.72 | 1.48 | 1.06 | 26 % | 41 % | 7.8e-5 | MVs$^{pIRESneo}$, EXOs$^{GnT-V}$, EXOs$^{pIRESneo}$ |
| Ubiquitin C-terminal hydrolase L5 *(UCHL5)* | 91.3 | 35.59 | 1.28 | 1.17 | 0 % | 53 % | 7.0e-4 | MVs$^{pIRESneo}$ |
| Fibromodulin *(FMOD)* | 19.0 | 15.85 | 1.28 | 0.95 | 15 % | 42 % | 3.5e-5 | MVs$^{GnT-V}$, MVs$^{GnT-III}$, EXOs$^{GnT-V}$, EXOs$^{GnT-III}$ |
| Ubiquitin conjugating enzyme E2 V2 *(UBE2V2)* | 36.9 | 29.39 | 2.25 | 1.08 | 0 % | 47 % | 5.8e-4 | EXO$^{GnT-V}$ |
| Hyaluronan and proteoglycan link protein 3 *(HAPLN3)* | 15.1 | 14.89 | 1.17 | 1.67 | 94 % | 21 % | 2.5e-4 | MVs$^{pIRESneo}$, MV$^{GnT-V}$, EXO$^{IRES}$, EXO$^{GnT-V}$ |

with Char-cot-Marie-Tooth disease type 4, a hereditary motor and sensory neuropathy. On the other hand, myosin IB (MYO1B) was implicated in only five pathologies, including melanoma, colorectal cancer, malignant neoplasm of the large intestine, hepatomegaly, and increased body weight. Based on the DisGenNET analysis, it appears that some of the identified EV proteins may have a better diagnostic potential in melanoma than others. For example, proteins assigned exclusively to melanoma can be used to distinguish melanoma from other potential malignancies that accompany or are confused with melanoma metastatic sites. Other identified proteins were assigned to a broader spectrum of neoplasms and carcinomas and therefore may be universal markers of cancer occurrence. Finally, even if proteins were assigned to numerous pathologies, it still increases their diagnostic potential beyond cancer-related pathologies.

In summary, the number and quality of available databases related not only to diseases, genes, and proteins, but also to glycoproteins, glycosylation patterns, and glycan structures, provide an incredible and growing scope for analyzing proteins and glycans contained in EVs. Such analysis can be performed not only in terms of their involvement in processes related to carcinogenesis (*e.g.* metastasis, angiogenesis, immune surveillance, migration, and dysregulation of cell cycle and cell death processes), but also their diagnostic and prognostic potential, and even the possibility of searching for potential therapeutic targets and ways to overcome drug resistance.

## 4. Discussion

The glycan structures synthesized by GnT-III and GnT-V are crucial for various aspects of melanoma biology. As previously shown, transfected WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$ cells display increased expression of the *MGAT3* or *MGAT5* genes, respectively [43], which was also confirmed in this study (Supplementary Data 1). Increased expression of proteins encoded by both genes and their increased activity in the respective transfectants were also demonstrated [43]. Furthermore, Link-Lenczowski et al. [119] used HILIC-HPLC coupled with MALDI-TOF and ESI-ion-trap-MS/MS to analyze the structures of 2-AA labelled N-glycans isolated from membrane and secretory proteins from WM266-4$^{GnT-III}$ cells. A variety of analyzed N-glycans were modified with bisecting GlcNAc residue as a result of GnT-III activity. However, highly branched complex structures binding strongly to PHA-L lectin were also observed, suggesting that increased GnT-III activity does not necessarily inhibit the formation of highly branched glycans (possibly by GnT-V).

It is well known that changes of the cellular glycosylation profile can be partially reflected within the EV glycome. Particularly, since EVs originate from cellular membranes, membrane glycoproteins would be the most important source of EV glycans. In this study, we analyzed by flow cytometry whether induced cellular *MGAT3* or *MGAT5* overexpression alters the surface glycosylation of melanoma-derived EVs. We observed the expected increase in the percentage of "bisecting" GlcNAc-expressing cells in PHA-E staining of WM266-4$^{GnT-III}$ cells, and

**Table 10**
Basic information on the glycosylation of proteins identified in EXOs and MVs derived from WM266-4$^{pIRESneo}$, WM266-4$^{GnT-V}$ and WM266-4$^{GnT-III}$ cells with prognostic value in melanoma patients. Data was obtained from the GlyConnect and GlyGen databases.

| Protein name | Gene | Reported glycosylation sites | Reported sites with N-glycans |
|---|---|---|---|
| Guanine nucleotide-binding protein G(i) subunit α-1 | *GNAI1* | 2 | Asn141 |
| Glutathione S-transferase Mu 3 | *GSTM3* | 1 | 0 |
| Insulin-like growth factor-binding protein 2 | *IGFBP2* | 3 | 0 |
| Protein disulfide isomerase family A member 6 | *PDIA6* | 1 | 0 |
| Selenium binding protein 1 | *SELP* | 1 | 0 |
| Biglycan | *BGN* | 5 | Asn270, Asn311 |
| Galectin 1 | *LGALS1* | 2 | Asn96 |
| Tetraspanin 9 | *TSPAN9* | 1 | Asn180 |
| Regulator of chromosome condensation 2 | *RCC2* | 2 | 0 |
| Methionine adenosyltransferase 2A | *MAT2A* | 3 | 0 |
| Cell division cycle associated 8 | *CDCA8* | 2 (predicted) | 0 |
| Malic enzyme 1 | *ME1* | 2 | 0 |
| Complement C1q binding protein | *C1QBP* | 2 | Asn114 |
| Cullin 3 | *CUL3* | 2 | Asn481 |
| Leucine zipper and CTNNBIP1 domain containing | *LCN2* | 1 | 0 |
| C1q and TNF related 5 | *C1QTNF5* | 2 | 0 |
| Ubiquitin C-terminal hydrolase L5 | *UCHL5* | N/A | N/A |
| Fibromodulin | *FMOD* | 4 | Asn166, Asn201, Asn291 |
| Ubiquitin conjugating enzyme E2 V2 | *UBE2V2* | 1 | 0 |
| Hyaluronan and proteoglycan link protein 3 | *HPLN3* | N/A | N/A |

**Table 11**
List of proteins common to individual EXO and MV populations derived from three genetic variants of WM266-4 cells and proteins present in melanoma pathways in the KEGG pathway and DISEASES pathway databases.
PMEL - melanocyte protein PMEL; GNA11 - guanine nucleotide-binding protein subunit alpha-11; GNAQ - guanine nucleotide-binding protein G(q) subunit alpha; NRAS - GTPase Nras, MTAP- S-methyl-5′-thioadenosine phosphorylase; HRAS - GTPase HRas; MAPK1 - mitogen-activated protein kinase 1; DDB- DNA damage-binding protein 2; PDGFA - platelet-derived growth factor subunit A; IGFR1 - insulin-like growth factor 1 receptor; MAPK1 - mitogen-activated protein kinase; MAP2K2 - dual specificity mitogen-activated protein kinase kinase 2; PDGFD-platelet-derived growth factor D; KRAS - GTPase Kras; AKT1 - RAC-alpha serine/threonine-protein kinase; MAP2K1 - dual specificity mitogen-activated protein kinase kinase 1; BAX - apoptosis regulator BAX; AKT3 - K RAC-gamma serine/threonine-protein kinase.

| Cell genetic variant | EV population | Proteins in DOID pathway (DOID:1909) | Proteins in KEGG pathway (hsa05218) |
|---|---|---|---|
| WM266-4$^{pIRESneo}$ | EXOs | PMEL | PDGFD |
| | MVs | GNA11, PMEL, GNAQ | HRAS, MAPK1 |
| WM266-4$^{GnT-V}$ | EXOs | GNA11, PMEL | DDB |
| | MVs | NRAS, GNA11, PMEL, GNAQ | NRAS, HRAS, PDGAFA, IGFR1, MAPK1, MAP2K2 |
| WM266-4$^{GnT-III}$ | EXOs | PMEL | PDGFD |
| | MVs | NRAS, GNA11, PMEL, GNAQ, MTAP | NRAS, GRAS, KRAS, IGFR1, MAPK1,AKT1, MAP2K2, MAP2K1, BAX, AKT3 |

the highest RFI for WM266-4$^{GnT-V}$ cells in PHA-L staining specific for β1,6-branched structures. Notably, *MGAT3* and *MGAT5* overexpression-related glycosylation changes were more dynamic on the vesicular level than on the cellular one. In PHA-E staining both WM266-4$^{GnT-III}$ and WM266-4$^{GnT-V}$-derived EXOs and MVs displayed an increased percentage of positive objects, whereas WM266-4$^{GnT-III}$-derived MVs showed the highest RFI in this staining. WM266-4$^{GnT-V}$-derived MVs had the highest percentage of positive objects in PHA-L staining. As for RFI, both MVs and EXOs from WM266-4$^{GnT-V}$ cells showed higher surface expression of β1,6-branched structures than EVs from WM266-4$^{GnT-III}$ cells. In general, surface glycosylation pattern of MVs was more similar to that of the transfected cells releasing them than that of EXOs. Most likely, it can be attributed to MV biogenesis mechanism involving their formation within the outer cell membrane.

Furthermore, the presented research was the first to study considering the glycosylation of EVs derived from different genetic variants of the same melanoma cell line. However, glycosylation of melanoma-derived EV was analyzed by our team before using a larger panel of cell lines (WM115, WM266-4, WM793, WM1205Lu) [33]. On the surface of MVs from all cell lines we observed abundant N-glycans with bisecting GlcNAc and β1,6-branched N-glycans using flow cytometry. Moreover, both percentages of positive EVs and RFIs were lower in PHA-L and PHA-E staining of WM266-4-derived MVs than WM115-derived MVs. WM115 is a less invasive (primary tumor-derived) isogenic counterpart of WM266-4 cell line, which is suggested by some degree of retention of possibly tumor-promoting glycoepitopes by more invasive melanoma cells. Finally, as revealed by PHA-E lectin blotting, WM266-4-derived MVs were enriched in glycans with bisecting GlcNAc in comparison to membrane fraction of parental cells. The same observation was not made for PHA-L staining.

In the light of the aforementioned results, it should be noted that surface N-glycosylation patterns of EVs are essential for transferring molecular information to recipient cells and can affect immune responses to EVs, EV uptake via endocytosis or phagocytosis, and ligand-receptor interactions. Changes in EVs surface glycosylation can occur if differently glycosylated proteins are transported to membrane regions involved in MV budding. Alternatively, membrane proteins can be transported by endosomes, which may become multivesicular bodies during EXO formation. Changes in EV protein glycosylation can also affect EV function after their uptake by recipient cell. Therefore, it is important to note that not only glycosylation of surface proteins is altered when the cellular expression of glycosyltransferases changes, but also the glycosylation of proteins present inside lipid bilayer of EV membrane.

To investigate this issue, we performed quantitative and qualitative LC-MS/MS proteomic analysis of EXOs and MVs derived from three genetic variants of WM266-4 cells. While a significant percentage of microvesicular (723 out of 1828, *i.e.,* 39.5 %) and exosomal (467 out of 936, *i.e.,* 49.9 %) proteins were identified regardless of the cellular origin of the sample (Fig. 3), some were identified as unique for specific genetic variants (from 5 protein unique for MVs derived from WM266-4$^{GnT-III}$ cells, up to 488 proteins unique for MVs derived from WM266-4$^{pIRESneo}$ cells). Interestingly, in the case of MVs, the highest number of 488 unique proteins was identified for WM266-4$^{pIRESneo}$ variant, suggesting that both *MGAT3* and *MGAT5* overexpression abolish protein sorting into MVs. In contrast, EXOs derived from WM266-4$^{GnT-V}$ cells had the highest number of unique proteins (168), indicating that proteins bearing β1,6-branched N-glycans were preferentially recruited to EXOs. The observed differences support the existence of a glycan-dependent protein sorting mechanism to EVs. However, it does not seem to be universal, but rather dependent on the EV population, or the molecular characteristics of the EV-releasing cell line.

This study was the first to describe changes in EV protein cargo in response to genetically altered glycosylation of EV releasing cells. However, GnT-III and GnT-V activity has already been analyzed for EV function. It was shown that breast cancer cells, characterized by reduced

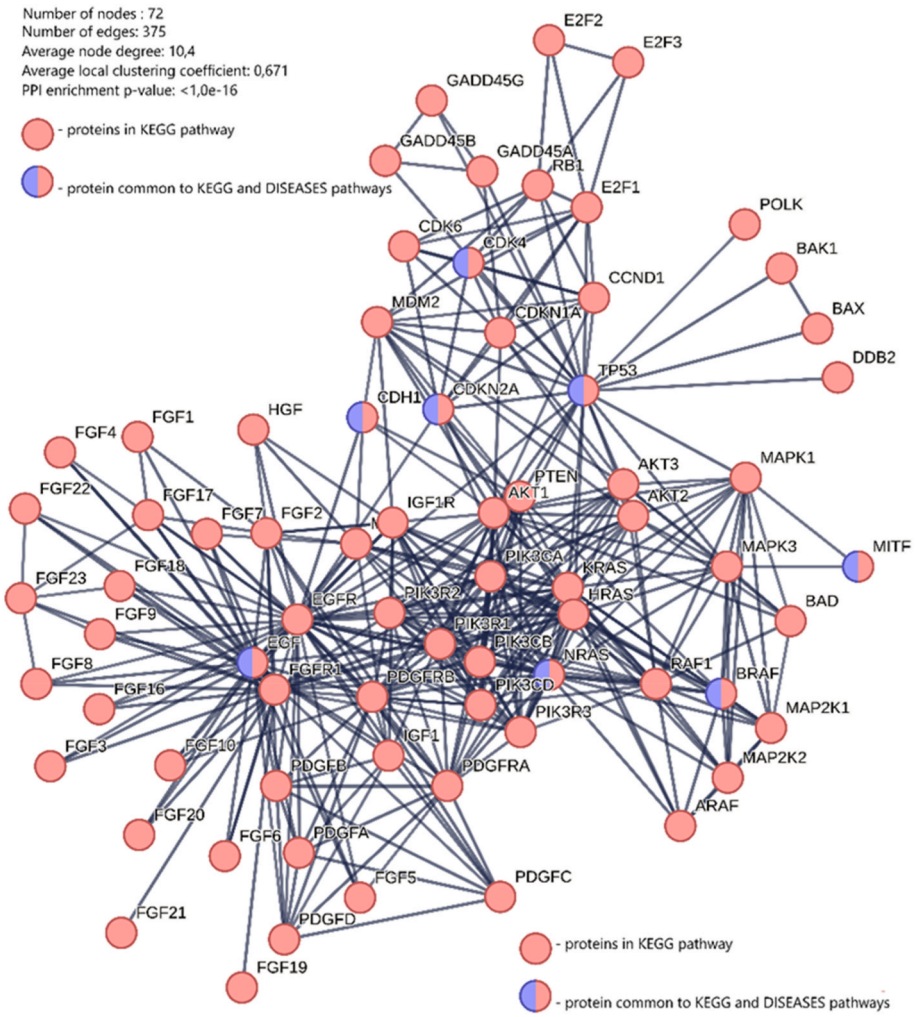



**Fig. 9.** Contribution of proteins expressed in melanoma EVs to the KEGG molecular pathway "Melanoma" (hsa05218) (red circles) and the pathway shared by KEGG and Disease Ontology DISEASES "Melanoma" (DOID:1909) pathways (blue and red circles). Interactome was prepared using https://string-db.org/v. 12.0. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

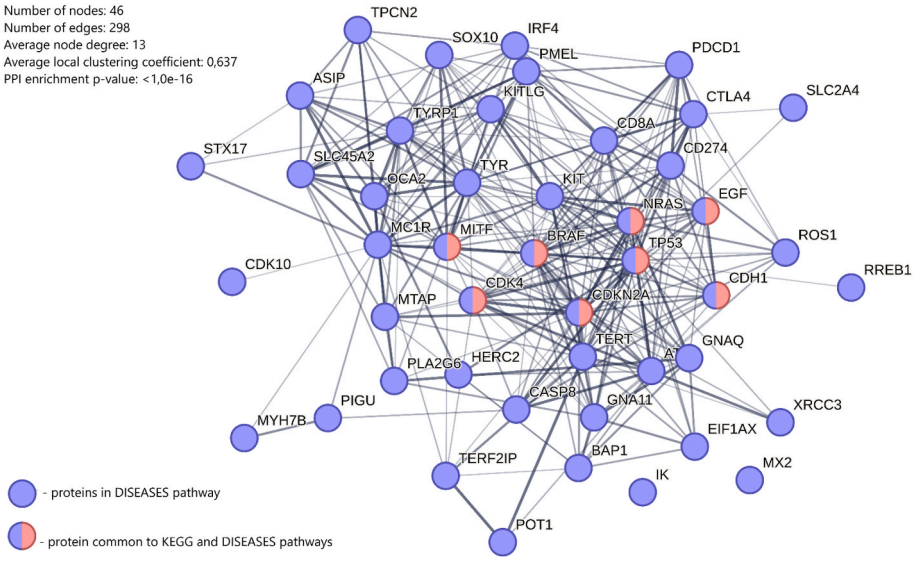



**Fig. 10.** Contribution of proteins expressed in melanoma EVs to the KEGG molecular pathway "Melanoma" (hsa05218) (red circles) and the pathway shared by KEGG and Disease Ontology DISEASES "Melanoma" (DOID:1909) pathways (blue and red circles). Interactome was prepared using https://string-db.org/v. 12.0. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

**Table 12**
List of genes encoding proteins present in EVs derived from WM266-4 melanoma cells compared with the DisGenNET database. Only proteins with DSI between 1 and 0,40 and available literature annotations for a role in melanoma are presented. The table shows the number of gene-involved diseases, gene diseases association (GDA) score, diseases specific index, diseases pleiotropy index, and chosen annotation context on protein/gene in melanoma. UBR4 - ubiquitin protein ligase E3 component N-recognin 4; PPP1R14B - protein phosphatase 1 regulatory inhibitor subunit 14B; MYO1B – myosin IB; RPS3 - ribosomal protein S3; MELTF – melanotransferrin; PPP6C - serine/threonine-protein phosphatase 6 catalytic subunit 6C; EPB41 – protein 4.1; QPCT - pituitary glutaminyl cyclase; ALDH1A3 - aldehyde dehydrogenase 1 family, member A3; CUL3 – cullin 3; PRDX2 - peroxiredoxin 2; MAP2K2 - mitogen-activated protein kinase 2; GLO1 - lactoylglutathione lyase; ASS1 - argininosuccinate synthase; NRDG1 - N-myc down-stream regulated 1; GNAQ11 - GNAQ-G protein subunit 11; MTAP - methylthioadenosine phosphorylase.

| Gene | Number of gene-involved diseases | Gene diseases association (GDA) | Diseases specific index | Diseases pleiotropy index | Chosen annotation context on protein/gene in melanoma | Literature |
|---|---|---|---|---|---|---|
| *UBR4* | 8 | 0.40 | 0.73 | 0.57 | Regulation of melanocytic differentiation phenotypes in melanoma. | [96] |
| *PPP1R14B* | 1 | 0.40 | 0.72 | 0.35 | Decreased expression inhibit cell cycle progression and proliferation rate in V3 versican isoform melanoma cells. | [97] |
| *MYO1B* | 5 | 0.40 | 0.66 | 0.65 | Depletion promotes apoptosis of melanoma cells. | [98] |
| *RPS3* | 2 | 0.50 | 0.65 | 0.74 | Regulates melanoma cell growth and apoptosis by targeting Cyto C/$Ca^{2+}$/MICU1 mitochondrial signalling; knockdown inhibited tumor growth *in vivo*. | [99] |
| *MELTF* | 1 | 0.75 | 0.63 | 0.65 | Attenuates melanoma progression in the A375-luc2 murine metastasis model and human patients. | [100] |
| *PPP6C* | 5 | 0.75 | 0.63 | 0.61 | Recurrently mutated has relationship with the melanocyte lineage and MITF; mutation in induce the development of melanoma. | [101, 102] |
| *EPB41* | 3 | 0.40 | 0.61 | 0.70 | Affects photodynamic therapy for B16 cells by regulating the transport of 5-aminolevulinic acid. | [103] |
| *QPCT* | 3 | 0.60 | 0.68 | 0.74 | Densely methylated in >95 % of uncultured melanoma tumor samples. | [104] |
| *ALDH1A3* | 10 | 0.60 | 0.55 | 0.4 | Upregulated by epigenetic mechanisms in melanoma cells compared with melanocytes. | [105] |
| *CUL3* | 15 | 0.40 | 0.53 | 0.91 | Identified as a mediator of vemurafenib resistance via Src-Rac1 signalling axis. | [106] |
| *PRDX2* | 14 | 0.65 | 0.52 | 0.91 | Significantly downregulated in melanoma compared to nevus; downregulation is related to increase production of ROS, proliferation, and migration; prevents metastasis by increasing E-cadherin/β-catenin complexes. | [107, 108] |
| *MAP2K2* | 27 | 0.75 | 0.5 | 0.91 | Mutated in ~8 % of melanoma patients; anti-CTLA-4 therapy might be more effective than anti-PD-1 therapy for patients with *MAP2K1/2*-mutated melanoma. | [109] |
| *GLO1* | 16 | 0.65 | 0.5 | 0.87 | Regulate detoxification and target adduction by methylglyoxal in malignant melanoma; deletion modulates glucose metabolism and redox homeostasis accelerating human A375 malignant melanoma tumor growth. | [110, 111] |
| *ASS1* | 14 | 0.9 | 0.49 | 0.87 | Reduced expression leads to arginine auxotrophy; ASS1 expression is the predominant mechanism for acquired resistance to ADI-PEG20 in cultured melanoma cells. | [112, 113] |
| *NDRG1* | 29 | 0.6 | 0.49 | 0.83 | Decrease the expression and activation of EGFR, HER2, and HER3 in response to EGF; upregulates by antitumor agents of the di-2-pyridylketone class. | [114] |
| *GNAQ11* | 14 | 0.9 | 0.47 | 0.87 | Mutated in non-uveal melanoma, rare subgroup of melanomas; frequently mutated in uveal melanoma. | [115, 116] |
| *MTAP* | 5 | 0.8 | 0.48 | 0.78 | Inverse correlation between MTAP protein expression and progression of melanocytic tumors; fusion with *ANRIL* gene results in epithelial-mesenchymal transition-like process. | [117, 118] |

bisecting GlcNAc due to low GnT-III activity, produce small EVs with low bisecting GlcNAc levels. Restoration of GnT-III activity in cells increased bisecting GlcNAc level in released EVs. Furthermore, EVs with low bisecting GlcNAc level increased carcinogenesis and metastatic potential in recipient cells, whereas those with high bisecting GlcNAc level reduced these pro-metastatic effects [120]. Another study also found that GnT-V activity was noticeably higher in EVs than in B16 melanoma, A549 lung carcinoma, and N2A neuroblastoma cells. GnT-V, enriched in EVs, was transferred to recipient cells and remodeled their N-glycan structures, suggesting that EVs support cancer-associated glycan changes [121].

GnT-V activity facilitates the expansion of N-glycans by the formation of multi-antennary structures, and the greater number of antennae in the sugar structure increases the possible number of terminations of antenna expansion by sialic acids. Sialylation of glycans is characteristic of hybrid-type but mainly complex-type N-glycans. Studies have shown that EVs derived from B16–F10 melanoma cells were more enriched in specific fractions of sialylated glycans obtained during anion-exchange chromatography than B16–F10 cells, and the total amount of N-glycans was about ten times greater in B16–F10-derived EVs. Despite this difference, N-glycan profiles remained consistent, suggesting that B16–F10 cells transfer their fundamental N-glycan structures and their relative abundances to B16–F10-derived EVs [122]. In the present study, however, we observed limited changes in the surface glycosylation of WM266-4 genetic variants and EVs released by these cells. In both MAA and SNA staining, overexpression of *MGAT3* and *MGAT5* resulted in a decreased percentage of positive cells, except SNA staining for WM266-4$^{GnT-V}$ cells. Regarding RFI, only WM266-4$^{GnT-V}$ displayed increased RFI in MAA staining. In the case of EVs, the highest percentage of MAA-positive vesicles and RFIs was observed for EXOs and MVs derived from WM266-4$^{GnT-III}$ cells. In SNA staining, EXOs and MVs derived from WM266-4$^{GnT-V}$ cells showed the highest percentage and RFI values, except for MVs derived from WM266-4$^{GnT-III}$ cells. Summing up, the effect of *MGAT3* and *MGAT5* overexpression on the surface sialylation pattern in melanoma cells seems to be cell-line dependent. Moreover, the expression of both α2,3- and α2,6-linked NeuNAc in EVs does not show a clear enrichment/depletion pattern compared to parental cells, suggesting more complex mechanisms linking GnT-III/GnT-V activity with subsequent sialylation of protein and their sorting into EVs.

LacNAc units are another possible extension of complex type N-glycans following β1,6-branching by GnT-V. Confocal microscopy study showed that poly/multiantenarry LacNAc glycans colocalize with CD63, glycosylated tetraspanin and EXOs marker, in membrane microdomains. Sk-Mel-5 cell and EV on blot staining using LacNAc-specific *Datura stramonium* (DSA) lectin and anti-CD63 antibody confirmed this overlap. Such observation was not made when cells were treated with deoxymannojirymycin (an inhibitor of complex type N-glycan synthesis), suggesting that EXOs originate from LacNAc-rich membrane regions, potentially guiding protein sorting [123]. Moreover, the presence of EWI-2, an N-glycosylated protein, in the presence of EXOs was significantly decreased when poly/multiantenarry LacNAc was removed by genetic deletion of glycosylation sites [123,124]. In the present study, we did not perform DSA lectin staining, but most likely LacNAc content of EXOs and MVs was changed, as action of GnT-III and GnT-V occurs before LacNAc is synthesized on a given glycan branch.

As mentioned, the present study is the first to use stably transfected cells with altered glycosylation to investigate its impact on protein cargo of released EVs. However, previously Harada et al. [125] used two glycosylation inhibitors, *i.e.* 2-deoxyglucose (2-DG), which integrates into N-glycan precursor (dolichol-linked oligosaccharide, DLO), causing its degradation and partial hypoglycosylation. Using label-free quantitative proteomics, they revealed that EVs derived from melanoma cells treated with 2-DG- or NGI-1 (another glycosylation inhibitor) showed depletion of cell surface receptors, hydrolases, transferases, transmembrane protein family members, and proteins involved in protein transport and melanin biosynthesis. On the other hand, N-glycosylation inhibition resulted in enrichment of EVs in proteins of non-exosomal but still vesicular origin (e.g., ATP citrate lyase, calreticulin and galectin 3 binding protein) but did not affect the abundance of proteins typical of exosome fractions (such as tetraspanins). Furthermore, site-specific N-glycosylation of the hepatocyte growth factor receptor has been shown to aid its sorting into EVs [125].

Finally, glycosylation of EVs, particularly of their surface proteins, is crucial for their biodistribution. There is substantial evidence that EV glycans facilitate their uptake by recipient cells or promote their accumulation in specific tissues or organs. For example, enzymatic removal of N- and O-glycans from EVs from BMD2a brain-metastatic breast cancer cells led to their increased uptake by endothelial cells. Furthermore, O-deglycosylation increased EV accumulation in the lungs, whereas N-deglycosylation had minimal effects on biodistribution, suggesting that surface glycosylation of cancer-derived EVs may limit cancer cell adherence to the lung and promote delivery to distant sites [126]. Another study showed that uptake of EXOs from adipose-derived stem cells is mediated by HeLa cell-surface-bound sialic acid-binding immunoglobulin (Ig)-like lectins (siglecs). The results were validated *in vivo* by observing the transport of EVs to lymph nodes and uptake by CD11b + antigen-presenting cells [127]. The involvement of NeuNAc in recipient cell-EV interactions was also confirmed in a study using neuraminidase-treated EVs from proliferative liver cells. They showed increased accumulation in the lung and better distribution to axillary lymph nodes compared to untreated EVs after injection into mice [128]. Similarly, EVs derived from murine hepatic cells AML12 cells treated with glycosidases (PNGase F and neuroaminidase) had enhanced uptake in skin, lung, retinal, ovarian, and prostate cell lines. Furthermore, desialylation reduced the uptake of EVs derived from another murine hepatic cell line MLP29 cells by NCI lung cancer and endothelial HUVEC [129]. These results indicate that N-glycans and their sialylation are crucial for targeting EV to organs potentially susceptible to metastasis [129]. However, all of the beforementioned studies focused on EV glycosylation, which was modified after vesicles isolation. Thus, future studies should investigate whether glycosylation changes induced in cells prior to EV release, such as overexpression of *MGAT3* or *MGAT 5* genes, exert similar effect on EV uptake and biodistribution.

One key area for further research is the role of glycosylation in protein sorting into EVs. Previous studies have focused on tracking of individual proteins, providing only fragmentary insights into the mechanisms governing their transport and selection. However, there has never been a comprehensive analysis of the entire EV proteome, including the glycosylation status of proteins, which limits our understanding of the role of glycans in EV protein sorting. Moreover, most previous studies have relied on the use of glycosylation inhibitors as tools to induce various glycan modifications on proteins. This approach has significant limitations. By the time an inhibitor is introduced, some glycans may already be present on proteins, making the effects of the inhibitor action only partial and difficult to control. In our approach, modification of glycosylation occurs at the genetic level, meaning that the process is altered from the very beginning. As a result, glycosylation is modified in a stable manner, allowing for a more precise study of its effects on the sorting of proteins into EVs and their biological function.

Moreover, different studies have used different approaches to identify melanoma biomarker proteins, with various alternative methods available. However, no comprehensive comparative studies have been conducted to systematically evaluate these different strategies. This knowledge gap limits our ability to determine the most effective approach to biomarker discovery. One potential direction is to move toward glycoproteomics. The challenge, however, is the vast number of proteins present in EVs, making data analysis very complex. A promising alternative is a glycomics-based approach, which can provide valuable insights into glycosylation patterns in EVs and their potential diagnostic significance based on analysis of EVs from melanoma patients with different stages of disease. In addition, targeted modifications of glycosylation pathways can be investigated by silencing or knocking out genes encoding specific glycosyltransferases or by using selective inhibitors. In summary, all of the above-mentioned strategies in the future would provide an opportunity to more accurately understand the impact of glycosylation on EV composition and the identification of reliable biomarkers of melanoma in EVs.

Looking ahead, in the next five years, this area of research may shift toward modifying the glycosylation of specific biomarker proteins and move to the clinical level. Once key biomarkers are identified, the focus can be on analysing glycosylation changes in a specific EV-associated protein at different stages of a given disease. This approach can provide a new layer of diagnostic precision. It is also important to note that EVs can be isolated from all physiological fluids, which offers a significant advantage. Sampling these fluids is much less invasive than traditional biopsies, making EV-based biomarker studies particularly promising for clinical applications.

## 5. Conclusions

The presented LC-MS/MS analysis identified a significant number of proteins in both EXOs and MVs with significant proteomic diversity between EVs derived from WM266-4 melanoma cells with induced overexpression of *MGAT3* and *MGAT5* genes. As the cellular glycosylation profile undergoes dynamic changes throughout melanoma progression, the EV cargo likely changes accordingly. Despite the potential of glycoproteomics to elucidate EV-recipient cell interactions and EV protein sorting mechanisms, studying these processes possess significant challenges. Glycans create diverse chemical barriers that obscure proteolytic cleavage sites, complicating mass spectrometry analysis of EV protein backbones. Furthermore, the diversity of glycan isomers binding to different glycosylation sites complicates the interpretation of MS data. Therefore, advanced computational tools, including high-performance networks, are required to navigate this complexity. In addition, the development of databases focusing on the expression of different mRNAs and proteins under specific conditions increases the accuracy of EV cargo analysis from both a diagnostic and prognostic perspective. Furthermore, glycosylation-related databases can provide key insights into the glycosylation status of proteins previously identified as biomarkers, allowing potential changes in their glycosylation to be considered in the diagnostic process. Further studies related to

melanoma glycobiomarkers present in EVs should focus on the correlation between EV biosynthesis and glycans, the impact of single protein glycosylation changes on their transport by EVs, the roles of different glycoforms of the same protein in EV-recipient cells interactions. Ideally, one should attempt to study individual glycoproteins, consistently present in both EXOs and MVs, regardless of tumor stage. This would allow tracking glycosylation changes that occur during melanoma progression. Understanding these changes is essential to fully develop the diagnostic and prognostic potential of the EV glycome.

## CRediT authorship contribution statement

**Magdalena Wilczak:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Magdalena Surman:** Writing – review & editing, Methodology, Investigation, Data curation, Conceptualization. **Urszula Jankowska:** Writing – review & editing, Methodology, Investigation, Data curation. **Bozena Skupien-Rabian:** Writing – review & editing, Methodology, Investigation, Data curation. **Małgorzata Przybyło:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

## Informed consent statement

Not applicable.

## Institutional review board statement

Not applicable.

## Data availability statement

The mass spectrometry data were deposited to the ProteomeXchange Consortium [130] via the MassIVE repository with the dataset identifier PXD057657.

## Funding

The research was funded by the BioS Priority Research Area under the program "Excellence Initiative – Research University" at the Jagiellonian University in Krakow (U1U/P03/NO/03.22).

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Abbreviations:

| | |
|---|---|
| α2M | α-2 macroglobulin |
| ApoE | Apolipoprotein E |
| BAX | Apoptosis regulator BAX |
| Cat D | Cathepsin D |
| CDK | Cyclin-dependent kinase |
| CTLA4 | Anti-cytotoxic T-lymphocyte associated protein 4 |
| DPI | Disease pleiotropy index |
| DSI | Disease specificity index |
| EGF | Epidermal growth factor |
| EGFR | Epidermal growth factor receptor |
| ESCRT | Endosomal sorting complexes required for transport |
| EVs | Extracellular Vesicles |
| EXOs | Exosomes |
| FPKM | \|Fragments per kilobase per million mapped fragments |
| GDA | Gene-disease association score |
| GlcNAc | N-acetylglucosamine |
| GnT-III | N-acetylglucosaminyltransferase III |
| GnT-V | β1,6-N-acetylglucosaminyltransferase V |
| GO | Gene Ontology |
| HSP | Heat shock protein |
| iBAQ | Intensity-Based Absolute Quantification |
| IGF1R | Insulin-like growth factor 1 receptor |
| LC-MS/MS | Liquid chromatography paired with tandem mass spectrometry |
| LDH | L-lactate dehydrogenase |
| LFQ | Label Free Quantitation |
| MAA | *Maackia amurensis* agglutinin |
| MAPK | Mitogen-activated protein kinase |
| MEK | Mitogen-activated protein kinase kinase |
| MELTF | Melanotransferrin |
| MHC | Major histocompatibility complex |
| MMP | Matrix metalloproteinase |
| MVs | Microvesicles |
| MYO1B | Myosin IB |
| NeuAc | N-Acetylneuraminic acid |
| NTA | Nanoparticle tracking analysis |
| nTPMM | Transcripts per million |
| OST | Oligosaccharyl transferase |
| PD-1 | Programmed death receptor 1 |
| PDGF | Platelet-derived growth factor |
| PHA-E | *Phaseolus vulgaris* erythroagglutinin |
| PHA-L | *Phaseolus vulgaris* leucoagglutinin |
| PMEL | Melanocyte protein PMEL |
| RFI | Relative fluorescence intensity |
| RNA | Ribonucleic acid |
| SNA | *Sambucus nigra* agglutinin |
| TACAs | Tumor-associated carbohydrate antigens |
| TCGA | The Cancer Genome Atlas |
| TEM | Transmission electron microscopy |
| TGF | Transforming growth factor |
| VEGFR1 | Vascular endothelial growth factor receptor 1 |

## Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2025.151749.

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